

The Gift of a Lifetime: The Hospital, Modern Medicine, and Mortality[†]

By ALEX HOLLINGSWORTH, KRZYSZTOF KARBOWNIK,
MELISSA A. THOMASSON, AND ANTHONY WRAY*

We explore how access to modern hospitals and medicine affects mortality by leveraging efforts of the Duke Endowment to modernize hospitals in the early twentieth century. The Endowment helped communities build and expand hospitals, obtain state-of-the-art medical technology, attract qualified medical personnel, and refine management practices. We find that Duke support increased the size and quality of the medical sector, fostering growth in not-for-profit hospitals and high-quality physicians. Duke funding reduced both infant mortality—with larger effects for Black infants than White infants—and long-run mortality. Finally, we find that communities aided by Duke benefited more from medical innovations. (JEL I11, I12, J13, J15, L31, N32, O31)

As medicine has advanced over the last century, spending on health care has risen to consume an ever-increasing share of resources. In 1900, health care represented only 2 percent of US gross domestic product (Craig 2006). By 2020, this share soared to 20 percent, with spending on hospital care alone accounting for 6 percent of US economic activity (CMS 2022). This past century has also marked the rise of modern medicine and hospitals, transforming the institution “from places of dreaded impurity and exiled human wreckage into awesome citadels of science and bureaucratic order” (Starr 2017, p. 145). Building on gains over the mid- to

*Hollingsworth: Department of Agricultural, Environmental, and Development Economics; Department of Economics; and John Glenn College of Public Affairs, the Ohio State University and NBER (email: hollingsworth.126@osu.edu); Karbownik: Department of Economics, Emory University and NBER (email: krzysztof.karbownik@emory.edu); Thomasson: Department of Economics, Miami University and NBER (email: thomasma@miamioh.edu); Wray: Department of Economics, University of Southern Denmark (email: wray@sam.sdu.dk). Erzo F.P. Luttmer was the coeditor for this article. We thank the editor and four anonymous referees whose comments have greatly improved the paper. We also thank Brian Beach, Brandyn Churchill, Jane Greve, Maggie Jones, Carl Kitchens, Peter Nencka, Greg Niemesh, and seminar/conference participants at the Atlanta-Athens Health Economists Research Conference, the Danish Society for Economic and Society History and Scandinavian Society of Economic and Social History annual meeting, the Essen Health Conference, the NBER Health Economics Fall Meeting, the NBER Summer Institute Development of American Economy and Children’s sessions, Duke University, Emory University, Indiana University, the Insper Institute of Education and Research, Lund University, Marquette University, Miami University, the Ohio State University, and the University of Copenhagen for helpful comments and feedback. We are grateful to Patrick Carlin and Mallory Dreyer for research assistance and Hardish Bindra at Paradigm Data Services. We would like to especially thank John Parman for sharing the North Carolina death certificate data. Thomasson acknowledges support from the Julian Lange Professorship. Wray appreciates financial support from the Japan Society for the Promotion of Science (KAKENHI Young Scientists B Grant J160100115), research funds from the Hitotsubashi Institute for Advanced Study at Hitotsubashi University, and an Arthur Cole Grant from the Economic History Association.

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late nineteenth century—attributable to rising incomes and better nutrition—health has improved dramatically over the twentieth century, with infant mortality rates falling by 95 percent and life expectancy at birth rising by 45 percent (Costa 2015). A large body of work has explored the drivers and the timing of this health transition. One claim from this literature is that modern medicine—with the exception of sulfa drugs—did not begin to materially affect health and mortality until after the 1950s (Cutler 2005). Despite this viewpoint, little is known about the origins of the modern hospital or the contribution of medical care in improving health in the first half of the twentieth century.

We seek to understand the role of funding for formal health care in improving health outcomes. Did financial resources help to expand hospitals and attract higher-quality physicians with modern medical training? Did these improvements to medical care contribute to declining mortality during infancy and in the long run? Given the many racial inequities in society and in health, were these effects similar or different by race? And finally, did the modernization of hospitals allow for more productive use of new medical technologies?

We leverage a combination of novel data and a unique quasi-experiment: a large-scale hospital modernization program introduced by the Duke Endowment in North Carolina during the first half of the twentieth century. Beginning in 1927, the Duke Endowment funded capital projects to upgrade hospitals, thereby expanding access to modern medicine and hospital care across the Carolinas. It did so by placing a particular emphasis on rural and Black residents—an uncommon approach in the Jim Crow-era South. The Endowment assisted hospitals in expanding and improving existing facilities, obtaining state-of-the-art medical technology, and elevating their management practices. In some communities, the Endowment helped to build new hospitals or to convert existing facilities to nonprofits. These modernization efforts were explicitly designed to attract higher-quality physicians to the Carolinas (Duke Endowment 1934; Durdin 1998). By the end of 1942, the Endowment had appropriated over \$3.3 million (\$53 million in 2017 dollars) for hospital expansion and modernization in North Carolina, an investment that significantly improved access to medical facilities and the quality of the medical infrastructure, thereby propelling the practice of medicine in supported communities toward the modern era.¹

Our main empirical approach is a difference-in-differences design that exploits quasi-random variation in Duke support, comparing counties that received a funding commitment from the Endowment to counties without such a commitment, and cohorts born before the initial commitment to those born after the introduction of Duke support. In this effort, we construct a novel dataset that combines information from hand-collected archival records, annual reports of the Duke Endowment, individual death certificates, physician directories, and a variety of other sources. We opt for an extensive-margin intent-to-treat specification as our preferred model because treatment was not solely monetary. To convert an appropriation into a payment, the Endowment typically required hospitals to undertake additional steps, such as changing accounting and record-keeping practices, ensuring that surgeries were conducted by licensed physicians, allowing *de facto* inspections by Endowment

¹ While the Duke Endowment operates in both North and South Carolina, we focus our analysis on North Carolina due to data availability, as we discuss in Section II.

employees, or forging community partnerships. An intent-to-treat analysis captures the full effect of this multifaceted treatment, allowing for Duke support to both impact health before a single dollar was received and facilitate changes that improved health even in hospitals that never received a payment.

We find that the efforts by Duke to modernize hospitals between 1927 and 1942 caused increases in hospital capacity and the quality of physicians. Duke support increased the number of hospital beds per 1,000 births in not-for-profit hospitals that were eligible for funding by 70.1 percent, while causing a decline in ineligible for-profit beds of 61.4 percent. Moreover, supported communities saw an increase in the number of high-quality physicians per 1,000 births of 60.2 percent and a decrease in the number of poorly trained physicians of 5.5 percent, thereby advancing the average state of medical knowledge in locations supported by Duke.

These first-stage effects and mechanisms help explain our core result that Duke financial support for the modernization of hospitals improved health, causing a 7.5 percent reduction in the infant mortality rate. Treatment effect estimates for Black infants (13.6 percent) are about three times as large as those for White infants (4.7 percent). Those exposed to Duke support around the time of birth also enjoyed lasting health effects, as their long-run mortality rates (between the ages 56 and 64) declined by 9.0 percent. Finally, we document that Duke-supported hospitals were able to use new medicines more effectively, suggesting a complementarity between high-quality hospitals and new medical innovations.

Our findings are robust. Point estimates and statistical significance are not materially affected by the choice of estimator, specification, transformation of the dependent variable, or sample decisions. Our findings are also insensitive to the use of recent techniques that are robust to treatment effect heterogeneity and staggered treatment adoption in a difference-in-differences setting. We further demonstrate that our results are not driven by the two primary threats to identification, differential pre-trends and selection into treatment. Across a number of event study specifications, outcomes in treated and control counties follow parallel trends prior to treatment. Using mortality data from other Southern states, we further confirm that the results are not driven by selection into treatment, by spillovers, or by the reallocation of medical expertise and investments within the Carolinas.

The program was cost-effective. Considering only infant mortality from 1927 to 1942, one life was saved for every US\$(2017)17,092 paid to hospitals, a cost significantly lower than any reasonable estimate for a value of statistical life (VSL). The intervention was also particularly effective in reducing Black infant mortality, resulting in a mortality rate decrease much larger than the estimated decrease for White infants (a reduction of 15.1 versus 3.1 deaths per 1,000 births) and narrowing the Black–White infant mortality gap by one-quarter. This new evidence that access to high-quality medical care and infrastructure disproportionately reduced Black mortality rates adds to previous work on the Black–White infant mortality gap, some of which finds that it has no relationship with increased health care access (Collins and Thomasson 2004; Elder, Goddeeris, and Haider 2016; Anderson, Charles, and Rees 2021; Anderson et al. 2021).²

²The Duke Endowment's Indenture of Trust, written in the 1920s, explicitly stated that hospital access should be provided for *both* Black and White individuals. Although Duke-supported hospitals remained segregated and

This paper makes contributions to several literatures. First, our findings contribute to a series of articles investigating the drivers of the health improvements observed in the early to mid-twentieth century. Most prior work focuses on the role of public health efforts (e.g., Anderson, Charles, and Rees 2022) or the advent of sulfa drugs (e.g., Thomasson and Treber 2008; Jayachandran, Lleras-Muney, and Smith 2010). Studies that examine the effects of hospitals or medical care on mortality have primarily concentrated on later time periods (Cutler 2005). For example, Figinski and Troland (2020) show that a rural hospital construction program in Appalachia in the 1950s improved infant health. Our results indicate that even earlier improvements in hospital construction and modernization reduced mortality, likely by attracting better-educated physicians to the area. Thus, our findings suggest that improvements in medical care may have played a greater role in reducing mortality in the first half of the twentieth century than has been acknowledged.

Second, we build on the broader literature examining the short- and long-run consequences of prenatal and infant health. We also complement studies examining the determinants of mortality and longevity, in which a common theme is that early life circumstances matter for later-life health (Almond, Currie, and Duque 2018). Despite this large literature, ours is the first study to leverage quasi-experimental variation to show both the short- and long-run mortality consequences of a large hospital-funding program, and to our knowledge, it is the first to examine whether hospital improvements in this time period helped to ameliorate long-standing racial gaps in mortality. A unique feature of our work is that we do not solely focus on applying a well-established intervention to a different vulnerable population. Instead, we analyze a bold experiment aimed at modernizing medical care during a critical phase in its development.

Third, we contribute to the health care literature by demonstrating that high-quality facilities can have complementary effects. We show that improving facilities attracted physicians who graduated from better medical schools. Such an outcome was a clear goal of the Duke Endowment, as the first director of the Endowment's Hospital Section, Dr. Watson S. Rankin, stated that "[t]he modern hospital . . . had come to play an essential part not only in the practice of medicine but also in the distribution of medical personnel" and that creating well-supported hospitals would be "the main remedy for the shortage of doctors" (Durdin 1998, pp. 33 and 12). Our findings support this view.

An additional complementarity that we examine is the relationship between technological innovation and improvements in physical and human capital, as discussed in Alsan and Goldin (2019). Specifically, we analyze the effect of the introduction of sulfa drugs in 1937, which resulted in significant reductions in infectious disease mortality (Thomasson and Treber 2008; Jayachandran, Lleras-Muney, and Smith 2010; Lazuka 2020). We demonstrate that Duke's emphasis on improving health care infrastructure and the quality of the medical workforce enabled communities to better leverage sulfa drugs, highlighting that the diffusion of knowledge is necessary

care may not have been equal across race within the same hospital, the Endowment extended hospital access to Black residents in ways that were far from cosmetic. Before the Endowment's biracial mandate, many Black residents had no access to hospital care, while afterward "all or practically all of the 65 hospital construction projects that the Endowment assisted during its first decade contained accommodations for both races" (Durdin 1998, p 86).

for the effective spread and use of medical technologies (Soares 2007). Our findings reveal that in communities with higher baseline infectious disease mortality rates, the impact of the Duke Endowment's support on infant mortality approximately *tripled* following the advent of sulfa drugs.

Fourth, we contribute to literature on management style, standardization, and firm performance (Bloom and Van Reenen 2007). A distinct aspect of Duke support is that it was a multifaceted bundle of treatments coupled with ample oversight that was not limited to a monetary component. In addition to providing funding for facilities and equipment, the Endowment altered core management practices and standardized hospital accounting procedures.³ It thereby “exerted considerable influence over the direction of hospital care” in supported communities (Gamble 1995, p. 116). Thus, it is unlikely that monetary assistance alone would have had the same effect. It is possible that the Duke intervention was so successful *because* these many changes served to reinforce one another.

Finally, our work connects with the literature examining the effects of charitable contributions on public health and welfare, including the hookworm eradication campaign sponsored by the Rockefeller Sanitary Commission (Bleakley 2007), and the Rosenwald Rural Schools Initiative (Aaronson and Mazumder 2011; Aaronson, Lange, and Mazumder 2014). For many years, the Duke Endowment was the third-largest charity in the United States, and it remains one of the largest charitable foundations in the world (Durdin 1998). Despite its size and success, to our knowledge, virtually no others (outside of historians documenting Endowment activities) have attempted to evaluate the effects of the many programs that it operated.

We view our results as having three important modern policy implications. First, the success of the Endowment's efforts to support hospitals was so pronounced that it “served as both inspiration and model” for the Hill-Burton Act of 1946 (Durdin 1998, p. 13), a large program designed to stimulate hospital construction and access across the United States (Chung, Gaynor, and Richards-Shubik 2017). Second, its impact in providing access to the first high-quality medical care facility in a community highlights the importance of attracting physicians to practice in underserved communities and the potential costs of closing rural hospitals today, as removing the last high-quality health care option may have a symmetric negative impact (Kozhimannil et al. 2018; Germack, Kandrack, and Martsolf 2019; Fischer, Royer, and White 2022). Third, our findings are relevant to health care investments in developing countries where donations from both private foundations and governments represent a major source of health financing (Yamey et al. 2019).

I. The Duke Endowment and Hospital Care in North Carolina

On December 11, 1924, James Buchanan Duke established the Duke Endowment with initial funding of \$40 million (\$563 million in 2017 dollars) to improve the social and economic welfare of communities in North and South Carolina. In the

³ Summarizing a letter written by Graham Davis, one-time assistant director of the Endowment, Durdin (1998, p. 38) notes that “the money they received from the Endowment was probably not as valuable to [the assisted hospitals] as the free service they got from the staff” in the form of consulting, oversight, management expertise, and assistance in converting to uniform record keeping, of whom Duke-supported facilities “were the largest group in the nation employing” (p. 35).

Indenture and Deed of Trust for the Endowment, Duke acknowledged the vital role of hospitals as “indispensable institutions” for “prolonging human life,” noting that “[t]he advance in the science of medicine” has made “hospital facilities essential for obtaining the best results in the practice of medicine” (Duke Endowment 1925, p. 11). Duke directed that 25.6 percent of the income from the invested Endowment funds be apportioned to the modernization, maintenance, and construction of not-for-profit hospitals (Duke Endowment 1925, pp. 10–11). Duke died unexpectedly in October 1925. His will left the Endowment with an additional \$69 million (\$972 million in 2017 dollars), with 90 percent of the annual income from this contribution specifically directed to hospitals in the Carolinas (Durdin 1998, p. 26).⁴

Another one of Duke’s stated objectives in improving hospital facilities was to attract more highly qualified doctors. In its inaugural report, the Hospital Section noted, “Hospitals hold and attract the better type of physicians . . . So it is that a local hospital builds up a profession, raises professional standards and serves to improve the practice of medicine not only for patients in the hospital but for patients in the whole county” (Duke Endowment 1925, pp. 144–45). This objective was realized in North Carolina, where physician density increased between 1923 and 1940 from 26 to 56 per 1,000 population. The share of births in hospitals likewise increased, rising between 1931 and 1942 from 8.9 percent to 34.2 percent, but remained low even compared to the average of Southern states (13.2 percent in 1931 and 42 percent in 1942).

Following Duke’s death and prior to disbursement of any funds from the Endowment, the foundation surveyed the health care landscape in the Carolinas with a focus on the quality, size, and density of hospitals, as well as the supply of physicians. Armed with this knowledge, the foundation started appropriating funds in 1927—the first year we consider a county to have been treated. Panel A of Figure 1 presents the share of North Carolina counties that received Duke appropriations by a specific year. No appropriations were made prior to 1927; however, by 1929, almost 30 percent of counties had been promised financial support. In subsequent years, the geographic spread slowed, with the share increasing by only an additional 20 percentage points between 1929 and 1942. At the time, the modal treated county-year had only one hospital. We present the spatial distribution of first appropriations by county and year in panel C of Figure 1.

Although Black residents composed 30 percent of the population in the Carolinas in 1924, they had far more limited access to hospitals than White residents due to institutionalized racial segregation. The Duke Endowment paid “special attention” (Thomas 2011, p. 78) to racial equality when making funding decisions and may have helped to narrow long-standing racial disparities in access to health care. In the period we study, the Endowment directed the vast majority of its support in North Carolina—75.5 percent of appropriations and 74.0 percent of payments—to hospitals that accepted patients of both races. White-only hospitals received 15.6 percent of capital appropriations (16.7 percent of payments), while Black-only hospitals received 8.9 percent (9.3 percent of payments).⁵

⁴ Separately in his will, Duke earmarked \$4 million (\$56 million in 2017 dollars) to “build and equip a medical school, hospital, and nurses’ home at Duke University” (Durdin 1998, p. 26).

⁵ While the Endowment paid “special attention [to] the needs of blacks in white-run biracial and black-run hospitals” (Thomas 2011, p. 78), Gamble (1995, p. 128) notes that this “support, came at a price . . . The assisted facilities would be ‘for blacks’ but would be supervised by whites.” Gamble (1997, p. 257) cites specific examples, including

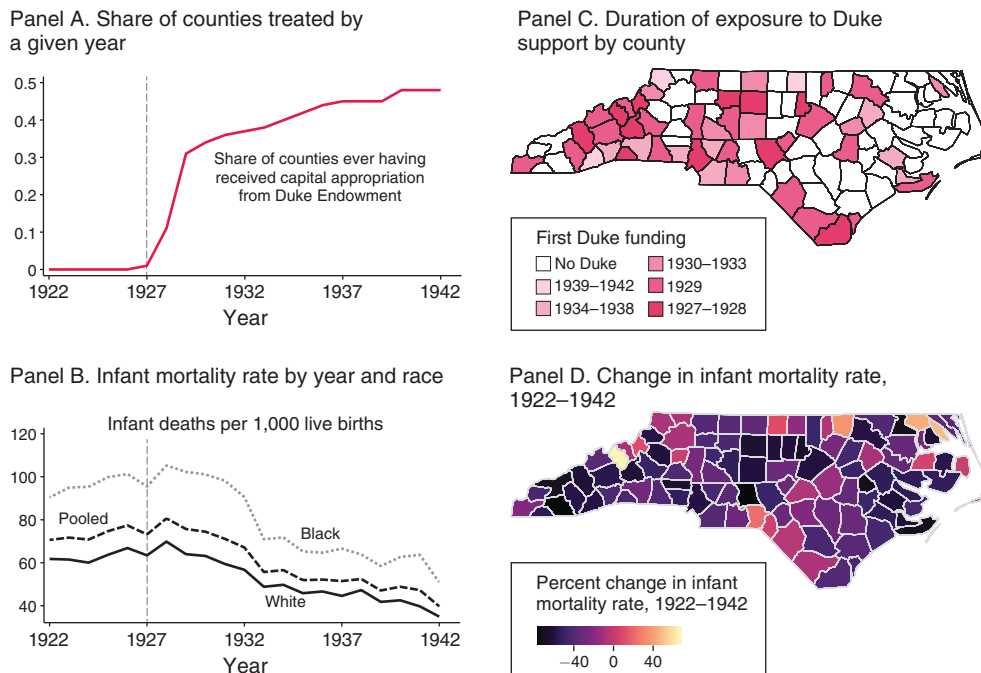


FIGURE 1. ROLLOUT OF DUKE SUPPORT AND CHANGES IN INFANT MORTALITY RATE, 1922–1942

Notes: Panel A plots the fraction of ever-treated counties in North Carolina by calendar year. Panel C shows a map of county boundaries for the 100 counties in the state of North Carolina. The color gradient provides a visualization of the time variation in the rollout of funding from the Duke Endowment. Counties are assigned to one of five groups based on the initial year in which they received funding (dark to light red): 1927–1928, 1929, 1930–1933, 1934–1938, 1939–1942. Darker counties were exposed to funding for a longer period of time. Counties that did not receive funding from the Duke Endowment during the sample period from 1922 to 1942 are colored white. Treatment is measured by appropriations for capital expenditures from the Duke Endowment. Treatment timing is based on the *Annual Report of the Hospital Section* for the years 1927 to 1942, published by the Duke Endowment. Panel B plots the average annual infant mortality rate per 1,000 live births in North Carolina between 1922 and 1942, weighted by county population, for Black infants (gray dotted line), White infants (solid black line), as well as Black and White infants pooled together (dashed black line). Panel D plots the percent change in the infant mortality rate per 1,000 live births in each North Carolina county between 1922 and 1942. Darker colors imply declines in infant mortality, while brighter colors imply increases in infant mortality.

II. Data

A. Hospital Funding

Our paper combines multiple hand-collected data sources into a county-by-year-of-birth panel. Our measures of treatment are derived from data on appropriations and payments for capital projects reported in the *Annual Reports of*

changes at Lincoln Hospital, where the Duke Endowment “limited the access of black physicians to the operating room and insisted that all their work be supervised by white surgeons” and replaced the Black superintendent with a White manager against the will of the surrounding Black community. Gamble (1997, p. 267) argues that this “arrangement frequently denied black people a voice in determining the direction of institutions which served them” and “reinforced a racial hierarchy within medicine.” She concludes that the Duke Endowment “contributed to the advancement and maintenance of a separate, and not equal, medical world for black physicians, nurses and patients” (Gamble 1997, p. 267).

the Hospital Section of the Duke Endowment (see online Appendix Figure A1 for an image of the raw data from the 1940 annual report). Each entry includes the name of the hospital, its location, as well as the dollar value of appropriations for capital expenditures, actual payments from the Endowment, and the purpose of the funding. An appropriation for capital expenditures represents a signed commitment from the Duke Endowment to reimburse costs associated with the expansion or improvement of existing hospitals as well as the construction of new facilities. Overall, 89.2 percent of appropriations were eventually paid. In most cases, the first payments on appropriations were delivered within two years (86.9 percent), and only 10.8 percent of appropriations never resulted in a payment (see online Appendix Figure E1).

Between 1927 and 1942, Duke approved appropriations for 130 projects at 65 hospitals in 48 counties, with 116 projects at 59 hospitals in 44 counties ultimately receiving funding by 1942. Appropriations are categorized by the “purpose” of a project. While these categories are not mutually exclusive (e.g., an addition could also include purchase of equipment), they are broadly informative of the types of activities funded by the Endowment. About 29.2 percent of projects approved for appropriations were for additions to existing not-for-profit hospitals, while purchases of equipment and of existing facilities (changing a for-profit facility into a not-for-profit) represented 22.3 percent and 10.0 percent of approved projects, respectively. Finally, 26.2 percent of approvals were for the construction of new hospital facilities, and 14.6 percent were for building homes for nurses (online Appendix Table E1).

B. Number of Hospitals and Hospital Beds

Our analysis of hospital capacity is based on data from “Hospital Service in the United States,” published in the *Journal of the American Medical Association* (American Medical Association 1942a; Esteves et al. 2022). These records include the count of hospitals and hospital beds in each county and are available for 1920, 1925, and annually from 1927 onward. To supplement these data, we include newly digitized hospital listings from the biannual *American Medical Directory* (AMD) (American Medical Association 1942b). We obtain additional data on hospitals serving Black patients from Pollitt (2017). We focus our analysis specifically on general hospitals, excluding tuberculosis sanitariums and specialty hospitals, which were not the focus of Duke funding. General hospitals are classified based on their control type, with our main focus on not-for-profit—including those designated as church run, nonprofit, or public—and proprietary (e.g., for-profit) hospitals. Not-for-profit status was a prerequisite for Duke support, and proprietary hospitals were required to convert to a nonproprietary status to access funding.

C. Number of Physicians by Race and Quality

In our estimates of the physician labor market, we use biannual data from the AMD covering the period 1921–1942, encompassing the universe of licensed physicians practicing in North Carolina. An individual entry includes the physician’s name, race, year of birth, place of practice, medical school, year of medical school graduation, and year of license (see online Appendix Figure A1 for an extract from

the AMD). We aggregate these data to the county-by-AMD-wave level. Following Moehling et al. (2020), we base physician quality on whether the medical school a physician graduated from required a two-year undergraduate degree at the time of their admission. As medical school was typically a four-year program, we define a doctor as high quality if they graduated at least four years after the medical school they attended introduced a two-year degree requirement as an admission prerequisite.⁶ All other doctors are considered low quality.

D. Births and Mortality

We focus on infant mortality in our analysis because we can accurately gauge exposure to Duke funding during infancy based on information about place and year of birth reported in individual North Carolina death certificates. In the absence of information on place of residence at death, we cannot reliably assign exposure to Duke funding in the case of adult mortality. Likewise, without longitudinal information on place of residence, we cannot accurately assign cumulative exposure to Duke funding over the life cycle.

Infant Mortality.—We obtain information on place of birth from the universe of individual North Carolina death certificates between 1917 and 1963 (Cook, Logan, and Parman 2014, 2016) and construct a new measure of infant mortality by county of birth based on the aggregation of individual deaths to the county-by-year-of-birth level (IPUMS USA 2015, 2021; US Geological Survey 2019).⁷ We define infant death as occurring between the day of birth and one year of life, limiting our sample to deaths of individuals born in North Carolina because similar infant mortality data are not available for South Carolina.

Births.—We obtain annual data on births by county, year, and race from 1922 to 1948 from volumes of the *Annual Report of the Bureau of Vital Statistics of the North Carolina State Board of Health* (North Carolina Bureau of Vital Statistics 1922). We use the number of births by year and county of occurrence to construct infant mortality rates and to use as weights in regressions with data at the county-by-year level. Data on births are not available prior to 1922.⁸ We combine these data with the count of infant deaths by county and year of birth to construct the infant mortality rate per 1,000 live births.

⁶Moehling et al. (2020) provide much greater discussion about reforms in US medical education. Even prior to the Flexner report in 1910, the American Medical Association (AMA) recognized the need for sweeping changes in medical education. To be rated high-quality by the AMA, schools had to require two years of undergraduate coursework, and thus, we adopt the same delineation in our paper.

⁷Data on deaths by place of *residence* are not reported in the aggregate data until 1932—well into the rollout of Duke funding. James M. Parrott, the Secretary and State Health Officer of North Carolina, recognized the problem with reporting aggregate deaths by place of occurrence in the 1933 *Annual Report of the Bureau of Vital Statistics*: “Deaths have been allocated to the place of legal residence of the deceased before death. Prior to 1932, deaths had been recorded by place of occurrence only. This gives a distorted rate for those counties that have medical centers within their borders and for the counties from which their patients are drawn” (North Carolina Bureau of Vital Statistics 1933, p.7). Ideally, we would observe information on place of residence at the time of birth, but to the best of our knowledge, these data do not exist.

⁸The annual reports were not published between 1917 and 1921. We transcribed the data on births by place of occurrence from the North Carolina Bureau of Vital Statistics because publicly available data do not report births separately by race until 1959 (Bailey et al. 2015).

Panel B of Figure 1 depicts North Carolina's infant mortality rate between 1922 and 1942 for both races, as well as separately for Black and White infants. Until about 1927, the first year with an appropriation from the Duke Endowment, there are stable (if not slightly increasing) mortality rates that are approximately 50 percent higher for Black as compared to White infants. Starting in 1928, infant mortality for both races begins to decline steadily before dropping precipitously between 1932 and 1933. This drop was not uniform across the state. Panel D of Figure 1 shows that between 1922 and 1942, some counties experienced declines in infant mortality of over 50 percent, but there were other counties that saw nontrivial increases.

Our main analysis of infant mortality uses county-by-year-of-birth panel data from 1922 to 1942. As explained above, 1922 is chosen as the first year since data on births by race are not provided for earlier years. The choice of 1942 as the endpoint for the sample is motivated by a number of factors. First, as panel A of Figure 1 illustrates, the rollout of counties receiving Duke funding proceeded rapidly upon establishment of the Endowment, while the post-1942 expansion (not depicted) saw relatively few additional counties gain access to funding. Nevertheless, we also collected data on Duke Endowment funding for the period up to 1962, and in Section VI we document that our infant mortality results are robust to extending the end year of the panel. Second, by ending the sample in 1942, we limit any possible influence that World War II may have on the analysis.⁹ Finally, this choice facilitates consistency between the short-run and long-run analyses.

Long-Run Mortality.—We face two primary challenges in estimating the effect of Duke Endowment funding on long-run mortality. First, given our focus on the 1922 to 1942 birth cohorts, we cannot observe every completed life-span, as some individuals born during these years are still alive. Second, to the best of our knowledge, publicly available aggregate data on mortality for the cohorts of interest do not extend to the present day. Thus, we turn to the public-use version of the Social Security Administration's Numerical Identification Files (Numident) from the National Archives and Records Administration (NARA) that contains records for every Social Security Number assigned to individuals with a verified death or who would have been over 110 years old by December 31, 2007 (Social Security Administration 2007).¹⁰ The data include the exact dates of birth and death and a place of birth text string. Since place of birth is truncated at the twentieth position, we use a crosswalk file that maps the text strings to a geocoded location and associated county of birth (Black et al. 2015). We collapse the Numident at the county- and year-of-birth-by-follow-up-year level, inferring age at death as the difference between year of birth and year of death. We link these data to exposure to Duke Endowment support at the birth-county-by-year-of-birth level.

Construction of the estimation sample for long-run mortality differs from the short-run analysis in two ways. First, we consider the mortality rates across a range

⁹After the United States entered World War II on December 8, 1941, Fort Bragg in North Carolina was one of the major military installations in the nation. The war effort could have affected both fertility and the health of infants directly through programs targeted at spouses of conscripts, such as the Emergency Maternal and Infant Care Program.

¹⁰To the best of our knowledge, the NARA Numident is the only publicly available data source that can be used to study long-run mortality in our context. While the Life-M data include linked mortality and natality records for North Carolina, for our sample cohorts it has more limited coverage and only includes deaths up to 1994.

of ages rather than just the first year of life. This means that an observation is at the birth-year by county-of-birth by follow-up-year level, where a follow-up year refers to the year in which we observe long-run mortality for each birth cohort. Stated differently, each observation is the mortality rate by single year of age for a given birth-county and birth-year cohort (Surveillance, Epidemiology, and End Results Program 2022).

For administrative reasons, the public-use Numident is incomplete prior to 1988 (Goldstein et al. 2021), which means that 1932 is the earliest year we are able to include in our long-run analysis.¹¹ Because we want to ensure that our analysis is not confounded by the effects of Medicare eligibility, we do not extend our sample beyond the 1941 birth cohort.¹² To avoid biasing our estimates due to differential survival probabilities by age, we use a balanced panel of follow-up ages for each county-by-year birth cohort, which implies that all cohorts are observed exactly 9 times at ages 56 to 64 during available follow-up years (e.g., the 1932 cohort is observed from 1988 to 1996). We discuss the econometric implications of these sample restrictions in Section V.

Pneumonia Mortality.—The North Carolina vital statistics annual reports also include data on pneumonia mortality by county and year. We use this to construct the exposure shares for our analysis of sulfa innovation. Specifically, we compute average pneumonia mortality per 100,000 in the population from 1922 to 1926—the five years prior to the initial rollout of Duke funding. County population is obtained from the National Historical Geographic Information System (Manson et al. 2022).

III. Methods

We use a difference-in-differences research design with staggered treatment adoption to estimate the first-stage effects of hospital funding from the Duke Endowment on hospital capacity and physician supply and quality. Additionally, we examine the impact of hospital modernization on infant and long-run mortality. We consider a county to be treated beginning in the first year that it received an appropriation for capital expenditures and all years thereafter.

First Stage.—We first consider if Duke support had an effect on the medical sector. We refer to these analyses as the first stage throughout, and we estimate them using a linear two-way fixed effects regression model. We prefer a linear model for the first stage since it more easily enables comparisons across different mutually exclusive groups (e.g., a comparison of the effect for the number of high-quality doctors versus the effect for the number of low-quality doctors will also be informative for the total number of physicians). Our first-stage effects are estimated using the following model:

$$(1) \ Y_{ct}^R = \beta_0 + \beta_1 \mathbf{1}\{t \geq \text{First appropriation year}\}_{ct} + \zeta_c + \eta_t + \Theta \mathbf{X}_{ct} + \varepsilon_{ct},$$

¹¹ Deaths at ages younger than 56 for the 1931 and earlier birth cohorts occurred prior to 1988, the earliest year that the public-use Numident is complete. Thus, any reported deaths from earlier cohorts would introduce survival bias.

¹² Changing life expectancy across time provides another justification for restricting the set of birth-year cohorts and follow-up ages. Life expectancy at birth does not surpass age 65 until after 1940 for White individuals and after 1970 for Black individuals (Haines 2008; Arias 2014; Arias et al. 2021). See online Appendix Figure F1.

where c indexes county and t indexes year—either the calendar year for hospitals or a wave of the *American Medical Directory* for physicians.¹³ For hospitals, our primary outcome of interest, Y_{ct}^R , is the number of hospital beds or hospitals in a county and calendar year. R indexes the hospital type, which can be either not-for-profit (encompassing church-run and public hospitals), proprietary, or the combination of the two categories. For physicians, Y_{ct}^R represents the total number of doctors or is disaggregated by quality (high versus low) and race (Black versus White). For the first stage, we also express the dependent variables as rates per 1,000 live births.¹⁴

The treatment variable is $\mathbf{1}\{t \geq \text{First appropriation year}\}_{ct}$, which takes the value of one for the first year that a county received an appropriation from the Duke Endowment and for all years thereafter. The treatment effect of interest is β_1 , which captures the change in the number of hospital beds, hospitals, or doctors of a particular type compared to the untreated counterfactual. This is an intent-to-treat effect, as capital appropriations from the Duke Endowment represented a commitment to provide financial resources that did not account for either the utilization (i.e., actual spending) or the allocation of funding (i.e., what the money was spent on).

A vector of control variables is denoted by $\mathbf{X}_{ct}$: the percent of the population that is illiterate, Black, other nonwhite race, or residing in urban areas; retail sales per capita, which helps to control for New Deal spending (Fishback, Horrace, and Kantor 2005); and an indicator for the presence of a county health department (Ferrell and Mead 1936; Hoehn-Velasco 2018, 2021).¹⁵ In a balance test that follows the approach presented by Pei, Pischke, and Schwandt (2019), we do not find a statistically significant relationship between changes in covariates and the treatment, easing concerns that selection or differential trends are driving our effects (see online Appendix Table G1).¹⁶ Most notably, Endowment support does not have an effect on our best measure of economic activity, retail sales per capita. This indicates that our findings are driven by changes in health care provision rather than general economic improvement resulting from hospital modernization. Nevertheless, to enhance precision and reduce residual variation, we include the vector of control variables described above in our preferred specification. Finally, all specifications include county (ζ) and calendar year or AMD wave (η) fixed effects. We cluster the standard errors (ε_{ct}) by county to account for correlated errors within a county. In our preferred specification, we weight observations by the number of births that occur in each county-year.

¹³The AMD was published every second year from 1921, except for a three-year gap between 1931 and 1934.

¹⁴We use counts of births (which are published annually) to avoid potential issues found in other papers that use population estimates that are interpolated between decennial census years (Arthi, Beach, and Hanlon. 2022).

¹⁵Another possible confounder may be the presence of health insurance coverage. However, over the years considered in our analysis, very few Americans had any type of health insurance coverage, especially so in the early years of our sample. While state-level data are not available, national data show that only 9 percent of the US population had coverage for hospital care in 1940 (Health Insurance Institute 1977). During our sample period, this type of coverage (later known as Blue Cross) was sponsored by hospitals. In other words, the existence of hospitals in North Carolina was necessary for people to have access to Blue Cross. Furthermore, these hospital prepayment plans were slower to grow and enroll subscribers until states enacted enabling legislation that exempted them from insurance reserve requirements (Thomasson 2002). North Carolina did not enact such legislation until 1943, after our main sample ends.

¹⁶The p -value on a joint F -test for significance in a regression of Duke support on the covariates is 0.192.

Event Studies.—We also present event studies of the following form:

$$(2) \quad Y_{ct}^R = \beta_0 + \sum_{j=-p}^{-2} \beta_j \mathbf{1}\{t - \text{First appropriation year}_{ct} = j\} \\ + \sum_{j=0}^q \beta_j \mathbf{1}\{t - \text{First appropriation year}_{ct} = j\} + \zeta_c + \eta_t + \varepsilon_{ct},$$

where j indicates the time period relative to the first appropriation year. The coefficients β_{-p} to β_{-2} capture trends prior to the start of the funding, while coefficients β_0 to β_q capture the effects of access to Duke Endowment resources.¹⁷ The omitted category is one year prior to the start of the treatment. Our event studies display the coefficients ± 6 for hospitals or ± 5 for physicians and omit more extreme values, which are nonetheless included in the estimation. We use county- and year-of-birth cohort size as weights and avoid additional controls for ease of interpretation and to follow recent recommendations in the literature (Roth et al. 2022). Standard errors (ε_{ct}) are clustered by county to account for correlated errors within a county.

Robust Estimators.—In Figures 2 and 3, we also implement a selection of the recent innovations that address the concerns that the linear two-way fixed effects regression model may be biased in the presence of staggered treatment timing or treatment effect heterogeneity (Goodman-Bacon 2021a; Roth et al. 2022): stacked regression (Cengiz et al. 2019), extended two-way fixed effects (Wooldridge 2021), and the Callaway and Sant’Anna (2021) estimator (see also online Appendixes B and C).¹⁸

Poisson Estimation.—Our preferred estimation method for the analysis of infant and later-life mortality is Poisson pseudo-maximum likelihood (Correia, Guimarães, and Zylkin 2020), under which coefficients of interest can be interpreted as semielasticities—the percent change in the outcome due to Duke funding exposure around the time of birth.¹⁹ We prefer Poisson estimation to OLS for the mortality analysis due to the presence of zeros in the dependent variable and because mortality risk increases dramatically across the age ranges we examine. Dependent variables containing zeros are explicitly modeled and handled well by the Poisson (Santos Silva and Tenreiro 2011), avoiding the need to manipulate the data and introduce additional issues (Mullahy and Norton 2022; Chen and Roth 2023).²⁰ Another feature of the Poisson is that it allows for a proportional analysis

¹⁷The most extreme negative coefficient, p , in the event studies is 18 for hospitals and 9 for doctors, while the most extreme positive coefficient, q , is 15 for hospitals and 7 for doctors.

¹⁸Our implementation of stacked regression creates treatment-timing-group-specific datasets that include counties first exposed to Duke Endowment funding in a particular year and “clean control” counties that are never treated in various estimation windows. We stack these datasets and saturate the county- and year-of-birth fixed effects in our DiD regressions and event study specifications with indicators for each of the stacked datasets.

¹⁹Throughout the paper, “Poisson” refers to the Poisson pseudo-maximum likelihood estimation method, rather than the assumption that the dependent variable has a Poisson distribution.

²⁰Along a similar line, we do not use Poisson estimation for the first-stage analysis because the first-stage sample includes counties that have no time variation in hospitals or physicians (e.g., an untreated county could always have 15 beds); the county fixed effect would cause a perfect prediction of this outcome, which would cause these observations to drop out of the estimation. In this case, we prefer the linear model for ease of interpretation due to the ability to retain these counties. The mortality treatment effect estimates are similar when estimated using a linear model.

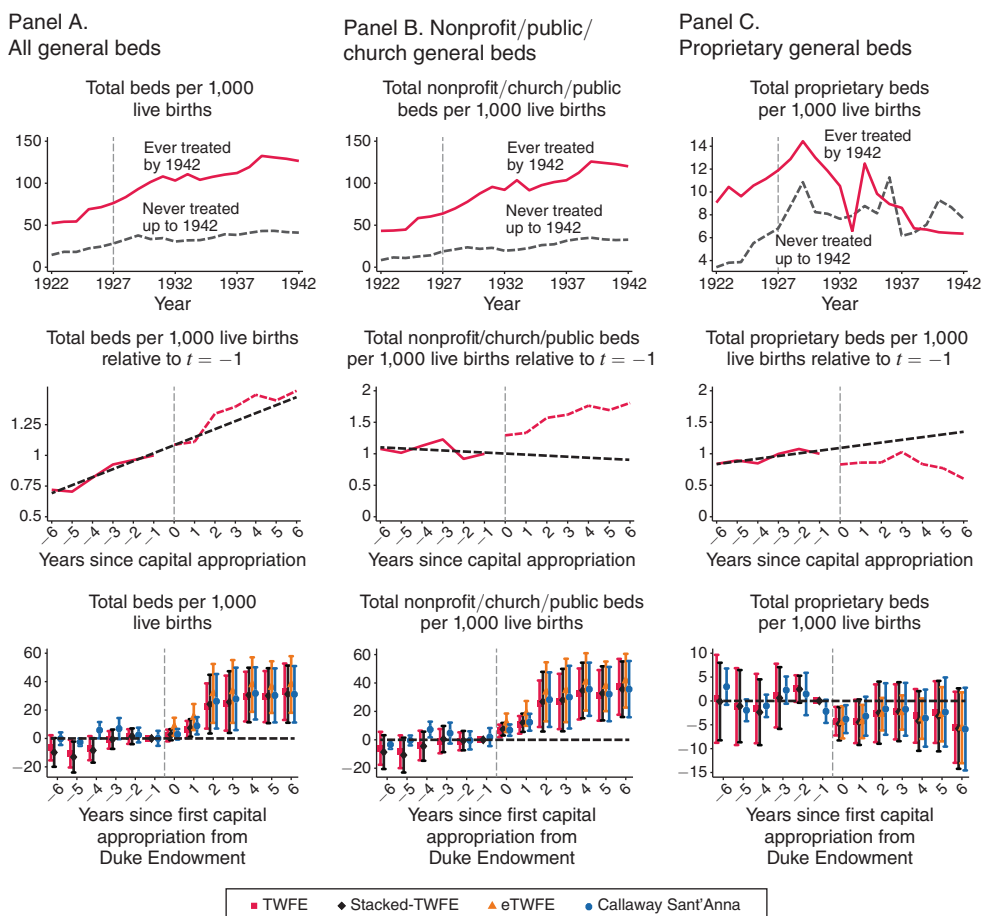


FIGURE 2. HOSPITAL BEDS AT THE COUNTY LEVEL BY TREATMENT STATUS, OWNERSHIP, AND EVENT TIME

Notes: Panel A plots results for the number of beds per 1,000 live births in all general hospitals, panel B plots the number of beds per 1,000 live births in nonprofit, public, or church-run general hospitals, and panel C plots the number of beds per 1,000 live births in proprietary general hospitals. The top row of plots show the average annual number of general hospital beds in a county by Duke treatment status. Counties “Ever treated by 1942” first received Duke funding during the sample period, between 1927 and 1942, while counties “Never treated up to 1942” did not. The middle row plots the number of general hospital beds relative to the year prior to treatment in counties treated by 1942. The black dashed line is a best-fit line estimated using only pretreatment data for treated counties. This trend is continued into posttreatment event time to serve as comparison to actual outcomes. The last row plots event study estimates of coefficient values and 95 percent confidence intervals for the lead and lag indicator variables for relative time periods from $t = -6$ to $t = 6$ around the first year that a county received an appropriation for capital expenditures from the Duke Endowment. More extreme relative time periods are estimated but not shown in the figures. The omitted category is -1 year before initial treatment. An observational unit is a county-by-year cell. Each plot shows four event study estimators: two-way fixed effects, stacked regression, extended two-way fixed effects (Wooldridge 2021), and Callaway and Sant’Anna (2021). All regressions include county and year fixed effects. Regressions are weighted by county-by-year-of-birth cohort size. Standard errors are clustered by county.

that ensures the comparability of estimates across groups with different baseline mortality risk. Thus, we can easily conduct a joint estimation of the long-run mortality effects without needing to account for the wide spread in baseline mortality risk across the follow-up ages we study.

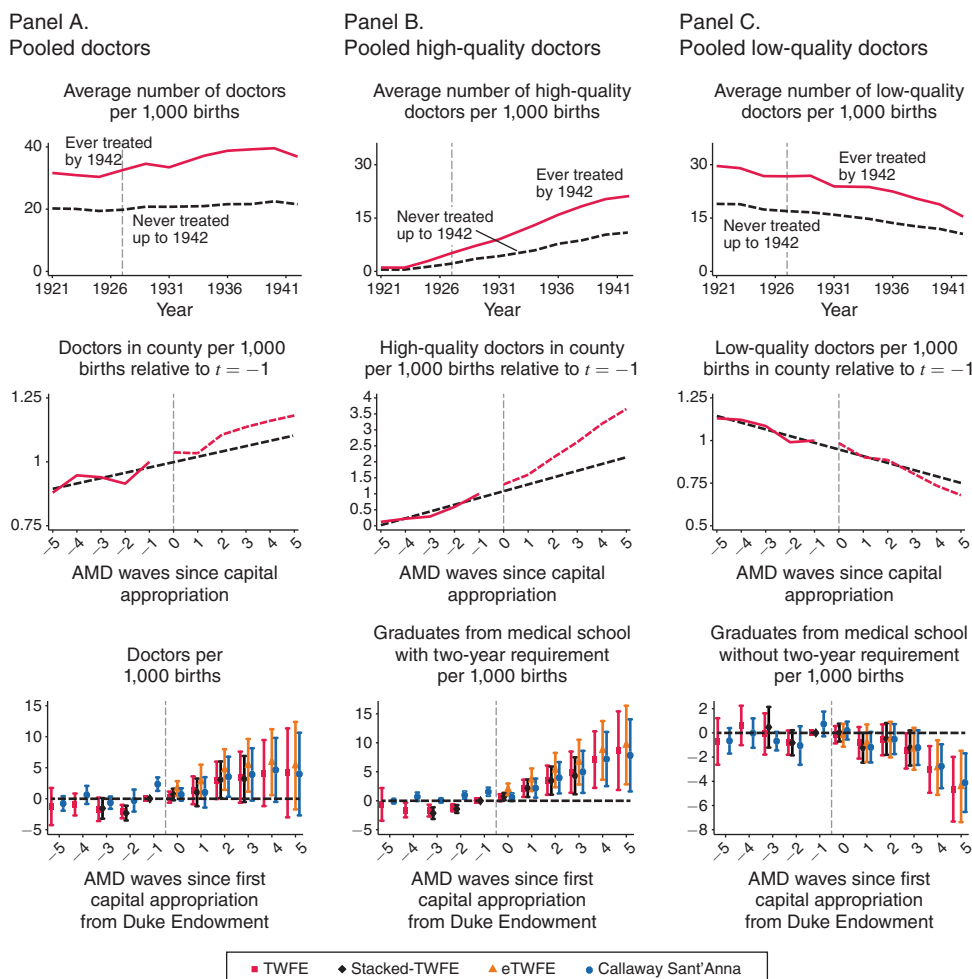


FIGURE 3. PHYSICIAN SUPPLY BY TREATMENT STATUS, QUALITY, AND EVENT TIME

Notes: Panel A plots results for the number of doctors, panel B plots the number of high-quality doctors, and panel C plots the number of low-quality doctors. All columns pool doctors by race. A high-quality doctor is one who graduated from a medical school at least four years after it introduced a two-year college degree prerequisite for admission. All other doctors are considered low quality. The top row of plots show the average annual number of doctors in a county by Duke treatment status. Counties “Ever treated by 1942” first received Duke funding during the sample period, between 1927 and 1942, while counties “Never treated up to 1942” did not. The middle row plots the number of doctors relative to the year prior to treatment in counties treated by 1942. Event time represents the number of *American Medical Directory* waves since the first year that a county received a capital appropriation from the Endowment. During the sample period, the AMD was published in 1921, 1923, 1925, 1927, 1929, 1931, 1934, 1936, 1938, 1940, and 1942. The black dashed line is a best-fit line estimated using only pretreatment data for treated counties. This trend is continued into posttreatment event time to serve as comparison to actual outcomes. The last row plots two-way fixed effects event study estimates of coefficient values and 95 percent confidence intervals for the lead and lag indicator variables for relative time periods from $t = -5$ to $t = 5$ around the first AMD wave after a county received an appropriation for capital expenditures from the Duke Endowment (or from $t = -3$ to $t = 3$ in the case of the stacked event study). More extreme relative time periods are estimated but not shown in the figures. The omitted category is -1 AMD wave before initial treatment. An observational unit is a county-by-AMD wave. Each plot shows four event study estimators: two-way fixed effects, stacked regression, the extended two-way fixed effects (Wooldridge 2021), and Callaway and Sant’Anna (2021). All regressions include county and AMD wave fixed effects. Regressions are weighted by county of birth cohort size averaged over the two or three years of the AMD wave. Standard errors are clustered by county.

Short-Run Mortality.—We estimate the following regression:

$$(3) \quad Y_{ct}^R = \exp[\alpha_0 + \alpha_1 \mathbf{1}\{t \geq \text{First appropriation year}\}_{ct} \\ + \zeta_c + \eta_t + \Theta \mathbf{X}_{ct}] \varepsilon_{ct},$$

where c indexes county of birth and t indexes birth year. Our primary outcome of interest Y_{ct}^R is the county-by-birth-year count of deaths or the infant mortality rate defined in Section IID. We consider both counts and rates as dependent variables to demonstrate that our construction of the denominator in the death rate does not drive our results. We also consider alternative definitions of the mortality variable, e.g., first-day infant mortality rate. We superscript this variable with R to indicate separate outcomes for Black and White persons or outcomes that pool together both racial groups.²¹ The treatment effect of interest is $\exp(\alpha_1) - 1$, which captures the percent change in infant mortality compared to the untreated counterfactual in our difference-in-differences framework. Given that the model is nonlinear, the treatment effect is identified under the assumption of parallel trends in the log of the expected potential outcomes (Lechner 2011). For the infant mortality analysis, we use a similar set of robust estimators as in the first-stage analysis, the differences being that we estimate stacked regression and extended two-way fixed effects by Poisson pseudo-maximum likelihood and log the dependent variable for Callaway and Sant’Anna (2021).

In our preferred Poisson specification, we add $\mathbf{X}_{ct}$, a vector of control variables described above, and weight observations by the number of births that occur in each county-year.²² Furthermore, all specifications include county-of-birth (ζ) and year-of-birth (η) fixed effects. We cluster the standard errors (ε_{ct}) by county of birth to account for correlated errors within a county.

We also present event studies of the following form:

$$(4) \quad Y_{ct}^R = \exp \left[\alpha_0 + \sum_{j=-18}^{-2} \alpha_j \mathbf{1}\{t - \text{First appropriation year}_{ct} = j\} \right. \\ \left. + \sum_{j=0}^{15} \alpha_j \mathbf{1}\{t - \text{First appropriation year}_{ct} = j\} + \zeta_c + \eta_t \right] \varepsilon_{ct},$$

where j indicates the time period relative to the first appropriation year. The coefficients α_{-18} to α_{-2} capture trends prior to the start of the funding, while coefficients α_0 to α_{15} capture the effects of access to Duke Endowment resources. The omitted category is one year prior to the start of the treatment. Our event study plots omit the

²¹ Hispanic people are not included as a separate racial category in our data. We do not have data on American Indians, who constituted only 0.5 percent of the North Carolina population at the time.

²² This method produces identical results to a model that uses the count of deaths as the outcome variable and includes an exposure variable equal to the number of births.

leftmost coefficients for relative time periods -18 to -7 , and the rightmost coefficients for relative time periods 7 to 15 . The dropoff in the number of treated counties by event time beyond ± 6 motivates the choice of these cutoffs (see online Appendix Figure I2). We weight the estimates by county- and year-of-birth cohort size and do not include additional controls. Standard errors (ε_{ct}) are clustered by county of birth.

IV. Results

A. First Stage: Hospitals

We first investigate the extent to which the Duke Endowment achieved James Buchanan Duke's vision to increase access to modern hospitals for the residents of North Carolina. We analyze the impact of support from the Duke Endowment on two key aspects of hospital capacity measured at the county-by-year level: total hospital beds and the number of hospitals. Given that the Duke Endowment explicitly restricted its funds to hospitals run on a not-for-profit basis, we separately analyze the effects for both proprietary (for-profit) and not-for-profit hospitals. This approach allows us to assess potential compositional changes in the hospital industry stemming from the Duke Endowment's support.

Figure 2 plots trends in hospital bed capacity in counties that received capital appropriations from the Duke Endowment by 1942 in comparison to those that did not (top row), and for treated counties, the ratio of beds per year to the number of beds in the year prior to treatment (middle row). Beds in nonprofit hospitals increased dramatically after the county received Duke funding, while beds in proprietary hospitals decreased, and the overall count increased as well.

Table 1 presents results based on equation (1) for the effects of Duke financial support on hospital capacity. Our simplest analyses, displayed in columns 1 to 3, report the effects on counts of beds (panel A) and hospitals (panel B), while columns 4 to 6 show the effects on beds and hospitals per 1,000 births. Across all specifications, results indicate that the Duke Endowment increased hospital capacity. In our preferred specification (reported in column 6) that focuses on beds per 1,000 births and includes both weights and controls, we find that Duke support increased the number of beds by 25 per 1,000 births, or 49.7 percent when calculated at the sample mean (see online Appendix Tables N1 and N2 for summary statistics). These increases were concentrated among beds in not-for-profit hospitals, which increased by 70.1 percent, compared to a decrease in for-profit beds of 61.4 percent. Event studies in Figure 2 validate these findings and generally confirm an absence of pre-trends. We extend these analyses to 1950 in Section L of the online Appendix and find that effects are persistent across the years included in this sample with a longer time frame as well.

Support from the Duke Endowment also resulted in an increase in the number of hospitals in North Carolina. While less precisely estimated across all specifications than the effects on beds, results reported in Table 1 suggest that Duke assistance resulted in an increase in not-for-profit hospitals by 0.33 per 1,000 births per treated county, or 36.3 percent. In contrast, proprietary hospitals decreased by 42.4 percent. At the extensive margin, only 40 of North Carolina's 100 counties had a general hospital in 1922. By 1942, 65 counties had at least 1.

TABLE 1—FIRST-STAGE EFFECTS OF DUKE SUPPORT ON HOSPITALS AND BEDS: EXTENSIVE-MARGIN INTENT-TO-TREAT ESTIMATES

	$Y_{ct}^R = \text{Beds or hospitals}$			$Y_{ct}^R = \text{Beds or hospitals per 1,000 births}$		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Beds</i>						
Total	35.46 (9.91)	44.72 (13.36)	32.63 (10.79)	31.11 (6.13)	27.48 (6.62)	24.57 (5.49)
Nonprofit/church/public	39.06 (10.10)	47.71 (14.10)	35.97 (11.21)	35.83 (6.26)	30.46 (6.90)	28.17 (5.86)
Proprietary	−4.63 (2.38)	−5.19 (3.24)	−5.72 (3.11)	−4.97 (2.63)	−4.35 (2.51)	−4.89 (2.59)
<i>Panel B. Hospitals</i>						
Total	0.18 (0.10)	0.11 (0.12)	0.10 (0.11)	0.31 (0.18)	0.15 (0.12)	0.20 (0.11)
Nonprofit/church/public	0.31 (0.10)	0.21 (0.14)	0.22 (0.12)	0.53 (0.16)	0.25 (0.11)	0.33 (0.12)
Proprietary	−0.14 (0.07)	−0.14 (0.08)	−0.17 (0.08)	−0.20 (0.12)	−0.12 (0.08)	−0.14 (0.08)
Observations	2,100	2,100	2,100	2,100	2,100	2,100
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Weights	No	Yes	Yes	No	Yes	Yes
Controls	No	No	Yes	No	No	Yes

Notes: Each coefficient comes from a separate two-way fixed effects regression. In columns 1 to 3, the dependent variable is the number of hospital beds (panel A) or the number of hospitals (panel B) in a county and year. Within panel A, the dependent variable is the number of beds in hospitals of any type (top row); in nonprofit, church-owned, or public hospitals (middle row); or in proprietary hospitals (bottom row). Panel B presents the same classification across rows with the number of hospitals of each type as the dependent variable. Columns 4 to 6 express the dependent variables as rates per 1,000 live births and follow the same structure as columns 1 to 3. Each coefficient represents the change in the outcome variable due to receiving a capital appropriation from the Duke Endowment. Control variables include percent illiterate, percent Black, percent other race, percent urban, retail sales per capita, and presence of a county health department. In columns 2 to 3 and 5 to 6, the weights are the number of births in a county and year. Standard errors are clustered at the county level.

B. First Stage: Physician Labor Market

Next, we consider whether the effects of support from the Duke Endowment extended beyond hospitals and into the physician labor market. Even though the funding was not directly tied to hiring physicians, those who ran the foundation believed that improving facilities would attract higher-quality doctors. The top row of Figure 3 shows the number of pooled, high-quality, and low-quality physicians per 1,000 live births across treatment status and calendar time. It is clear from this figure that the number of high-quality physicians increased in *both* treated and control counties and that the increase was greater in treated counties following the roll-out of support from the Endowment. Similarly, the number of low-quality physicians decreased in both treated and control counties, with a somewhat greater decrease in treated counties.

Table 2 presents the physician labor supply results. We document the effects on the number of doctors (columns 1 to 3) and doctors per 1,000 births (columns 4 to 6) measured at the county-by-AMD-wave level for all physicians (panel A), as well

TABLE 2—FIRST-STAGE EFFECTS OF DUKE SUPPORT ON NUMBER OF DOCTORS: EXTENSIVE-MARGIN INTENT-TO-TREAT ESTIMATES

	$\gamma_{ct}^R = \text{Doctors}$			$\gamma_{ct}^R = \text{Doctors per 1,000 live births}$		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Pooled</i>						
All	4.63 (2.08)	8.06 (3.26)	5.64 (2.47)	1.28 (1.29)	3.40 (1.44)	2.83 (1.23)
High quality	7.87 (2.32)	12.22 (3.45)	9.02 (2.58)	3.05 (1.22)	4.80 (1.44)	3.88 (1.23)
Low quality	−3.13 (1.00)	−4.16 (1.34)	−3.35 (1.11)	−1.70 (0.73)	−1.40 (0.73)	−1.02 (0.69)
Observations	1,100	1,100	1,100	1,100	1,100	1,100
<i>Panel B. Black</i>						
All	0.41 (0.21)	0.79 (0.44)	0.57 (0.31)	0.02 (0.90)	1.17 (0.65)	0.80 (0.57)
High quality	0.74 (0.24)	1.60 (0.44)	1.34 (0.37)	1.36 (0.47)	2.53 (0.67)	2.31 (0.59)
Low quality	−0.33 (0.15)	−0.81 (0.27)	−0.77 (0.22)	−1.34 (0.86)	−1.37 (0.51)	−1.51 (0.55)
Observations	1,078	1,078	1,078	1,078	1,078	1,078
<i>Panel C. White</i>						
All	4.21 (1.93)	6.77 (2.80)	4.38 (1.98)	−0.52 (2.40)	3.60 (2.09)	2.64 (1.72)
High quality	6.97 (2.10)	9.87 (2.96)	6.71 (2.00)	1.58 (2.20)	4.25 (1.99)	2.79 (1.61)
Low quality	−2.65 (0.94)	−3.11 (1.36)	−2.31 (1.07)	−1.91 (1.14)	−0.67 (1.04)	−0.11 (0.94)
Observations	1,078	1,078	1,078	1,078	1,078	1,078
County FE	Yes	Yes	Yes	Yes	Yes	Yes
AMD Wave FE	Yes	Yes	Yes	Yes	Yes	Yes
Weights	No	Yes	Yes	No	Yes	Yes
Controls	No	No	Yes	No	No	Yes

Notes: Each coefficient comes from a separate two-way fixed effects regression. In columns 1 to 3, the dependent variable is the number of doctors (panel A), the number of Black doctors (panel B), or the number of White doctors (panel C) in a county and publication year of the *American Medical Directory* (AMD wave). During the sample period, the AMD was published in 1921, 1923, 1925, 1927, 1929, 1931, 1934, 1936, 1938, 1940, and 1942. Within each panel, the dependent variable is the number of doctors (top row), the number of high-quality doctors (middle row), or the number of low-quality doctors (bottom row). A high-quality doctor is one who graduated from a medical school at least four years after it introduced a two-year college degree prerequisite for admission. All other doctors are considered low quality. Columns 4 to 6 express the dependent variables as rates per 1,000 live births and follow the same structure as columns 1 to 3. Each coefficient represents the change in the outcome variable due to receiving a capital appropriation from the Duke Endowment. Control variables include percent illiterate, percent Black, percent other race, percent urban, retail sales per capita, and presence of a county health department. In columns 2 to 3 and 5 to 6, the weights are the average number of births in each county for the years in the AMD wave. Panels B and C drop counties that ever have zero race-specific births between 1922 and 1942 for any AMD wave. Standard errors are clustered at the county level.

as for Black (panel B) and White (panel C) physicians. Within each panel, we also measure the effects of Duke support on high- and low-quality physicians. Column 1 represents the simplest specification, where weights and controls are not included and the dependent variable is the count of physicians. Our preferred specification in column 6 expressed as a rate includes weights and controls, and it is equivalent to the specification we later use to examine the impact of Duke support on the infant mortality rate.

In the pooled sample, the results indicate that the Duke Endowment realized its goal of attracting better-qualified physicians to the hospitals it supported. Table 2 shows a statistically significant positive effect of Duke Endowment support on the total labor supply of physicians in a county and a shift in the composition of physicians—an increase in higher-quality physicians and a decrease in lower-quality ones. The results, reported in our preferred specification in column 6, suggest that the number of doctors per 1,000 live births increased nearly 11.1 percent, with high-quality doctors increasing by 60.2 percent and low-quality physicians decreasing by 5.5 percent, although the latter is less precisely estimated. This change in physician composition likely had meaningful effects on the state of medical knowledge in an area. In our sample, “higher-quality” physicians were licensed, on average, 21 years later than “lower-quality” physicians. This means that for every low-quality physician replaced by a newer one, the state of medical knowledge advanced by more than two decades. When estimating the impact of Duke support by the race of physician, we observe a 245.7 percent increase in the number of high-quality Black physicians (2.3 per 1,000 births), and of high-quality White physicians by 28.0 percent (2.8 per 1,000 births). Conversely, low-quality Black physicians decreased by 50.7 percent, but the number of low-quality White physicians per 1,000 births was less affected.

We also explore alternative measures of physician quality, considering factors such as whether the physician’s medical school ever received an A or A+ rating from the American Medical Association (AMA), its existence and approval status by the AMA in 1942, or if it had closed or been absorbed by another medical school between 1922 and 1942, and find similar results (see online Appendix Tables C2 and C3). While most physicians were generalists in the early twentieth century, we find that Duke funding led to an increase in surgeons and specialists, as defined by various measures of specialization, including AMA membership, fellow status, and society memberships (see online Appendix Table C4). We also find suggestive evidence from decennial US census data that Duke funding increased employment in the medical and hospital industry among nurses, attendants, and clerical workers (see online Appendix Figure C3). These figures provide evidence that skilled hospital employment, including the clerical workers needed to implement the uniform accounting and record-keeping systems championed by the Duke Endowment, likely increased alongside Duke support.

C. Infant Mortality

We begin by presenting raw infant mortality data by calendar year and treatment status. The first row of Figure 4 depicts infant mortality rates per 1,000 live births in North Carolina between 1922 and 1942 separately for counties that ever received Duke Endowment support by 1942 (solid pink line) and those that did not (dashed black line). We present results pooled across racial groups (panel A) and separately for Black (panel B) and White infants (panel C). There are three descriptive patterns of interest in these figures. First, before the initial appropriation was granted in 1927, the infant mortality rates trended in a similar way in both treated and untreated counties, suggesting that our identifying assumption of parallel trends is valid. Second, the infant mortality rate is higher in the pre-Endowment period for eventually treated

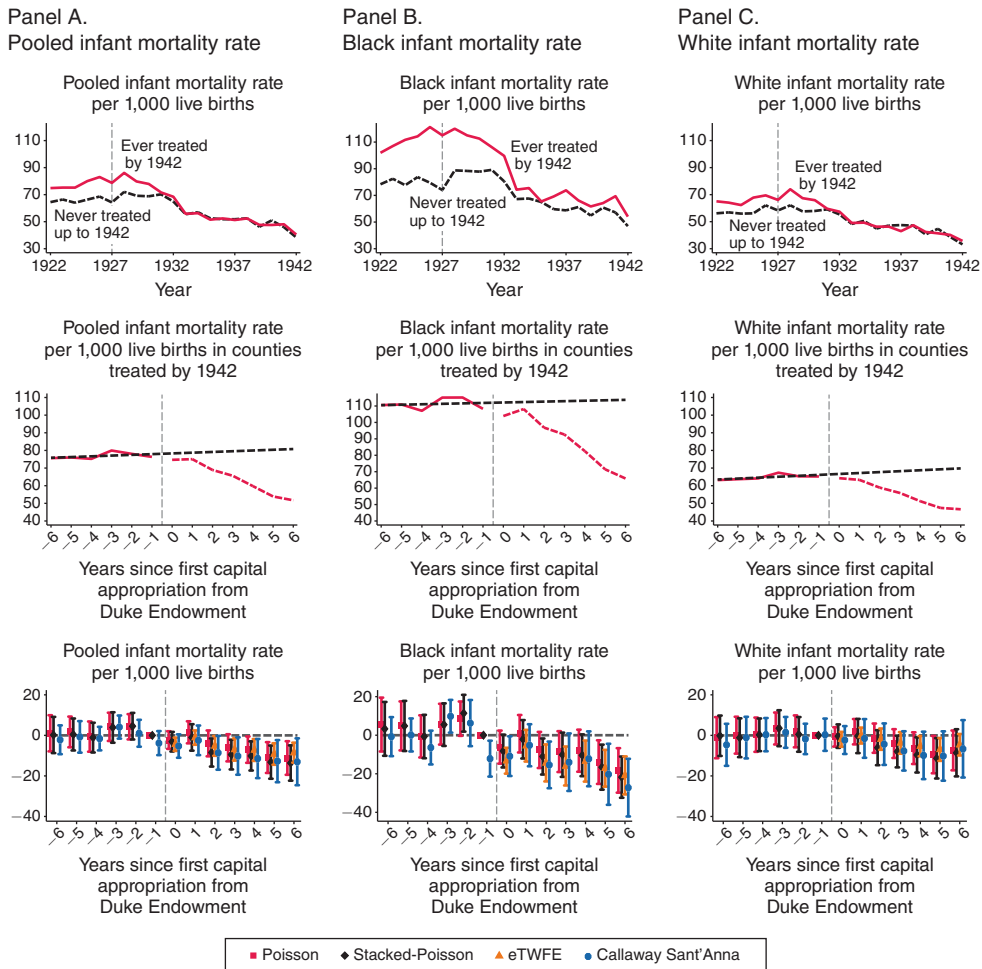


FIGURE 4. EVENT STUDY ESTIMATES AND INFANT MORTALITY RATES BY TREATMENT STATUS, RACE, AND EVENT TIME

Notes: The top row of plots show the annual infant mortality rate per 1,000 live births by Duke treatment status for Blacks and Whites pooled together, Blacks only, and Whites only, respectively. Counties “Ever treated by 1942” first received Duke funding during the sample period, between 1927 and 1942, while counties “Never treated up to 1942” did not. The middle row plots the infant mortality rate per 1,000 live births in counties treated by 1942 in event time relative to the first year of receiving a capital appropriation from the Endowment for the same three respective groups as in the top row. The black dashed line is a best-fit line estimated using only pretreatment data for treated counties. This trend is continued into posttreatment event time to serve as comparison to actual outcomes. The last row plots Poisson pseudo-maximum likelihood regression estimates of exponentiated coefficient values and 95 percent confidence intervals for the lead and lag indicator variables for time periods from $t = -6$ to $t = 6$ around the first year that a county received an appropriation for capital expenditures from the Duke Endowment. Our event study plots omit the leftmost coefficients for relative time periods between -18 and -7 and rightmost coefficients for relative time periods between 7 and 15 . The omitted category is -1 year before initial treatment. An observational unit is a county-by-year birth cohort. Coefficients reported in the table are transformed in the following way: $100 \times [\exp(\beta) - 1]$. The dependent variables are the infant mortality rate per 1,000 live birth in birth county c and year t : pooled by race, for Blacks only, and for Whites only, respectively. All regressions include county and year fixed effects. Regressions are weighted by county-by-year-of-birth cohort size. Standard errors are estimated using the delta method and are clustered by county of birth.

TABLE 3—EFFECT OF DUKE SUPPORT ON INFANT DEATHS AND INFANT MORTALITY RATE: EXTENSIVE-MARGIN INTENT-TO-TREAT ESTIMATES

	Y_{ct}^R = Infant deaths			Y_{ct}^R = Infant mortality rate		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Pooled infant deaths or infant mortality rate</i>						
Percent effect from Duke (= 1)	−6.12 (2.34)	−6.05 (2.56)	−6.51 (2.47)	−8.09 (2.51)	−8.11 (2.36)	−7.49 (2.30)
Observations	2,100	2,100	2,100	2,100	2,100	2,100
<i>Panel B. Black infant deaths or infant mortality rate</i>						
Percent effect from Duke (= 1)	−9.30 (3.52)	−8.50 (3.54)	−11.37 (3.29)	−7.97 (7.59)	−12.15 (3.46)	−13.61 (3.19)
Observations	1,995	1,995	1,995	1,995	1,995	1,995
<i>Panel C. White infant deaths or infant mortality rate</i>						
Percent effect from Duke (= 1)	−4.44 (2.94)	−4.25 (3.73)	−4.14 (3.02)	−8.00 (3.49)	−5.95 (2.98)	−4.71 (2.93)
Observations	1,995	1,995	1,995	1,995	1,995	1,995
<i>p</i> -value for difference by race	0.25	0.40	0.09	1.00	0.13	0.02
County of birth fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year of birth fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Weights	No	Yes	Yes	No	Yes	Yes
Controls	No	No	Yes	No	No	Yes

Notes: Each coefficient comes from a separate Poisson pseudo-maximum likelihood regression. In columns 1 to 3, the dependent variable is the number of infant deaths in a county and year. In columns 4 to 6, it is the infant mortality rate per 1,000 live births. Each coefficient represents the percent reduction in infant mortality due to receiving a capital appropriation from the Duke Endowment. A coefficient of -10 would mean that infant mortality declines by 10 percent. Coefficients reported in the table are transformed in the following way: $100 \times [\exp(\beta) - 1]$. Control variables include percent illiterate, percent Black, percent other race, percent urban, retail sales per capita, and presence of a county health department. In columns 2 to 3 and 5 to 6, the weights are the number of births in a county and year. Panels B and C drop counties that ever have zero race-specific births between 1922 and 1942. The bottom row of panel C presents a *p*-value from the interaction of race with our treatment variable from a model that fully interacts all variables with race. Standard errors are estimated using the delta method and are clustered at the county level.

counties, which is consistent with the idea that the Endowment targeted worse-off and underserved communities. Third, after 1927, the infant mortality rate in treated counties converged to that of untreated counties, suggesting that the Endowment could have played a causal role in driving the observed data patterns.

The middle row of Figure 4 plots average infant mortality for treated counties in event time. Pretreatment data are displayed using a solid pink line, and posttreatment data are plotted by a dashed pink line. The dashed black line shows a best-fit line estimated using only pretreatment data. This trend is continued into the posttreatment period, providing a straightforward basis for comparisons of potential outcomes. These figures show a clear trend break following the first capital appropriation from the Endowment. In each figure, the deviation is substantial and negative, indicating that Duke support likely reduced infant mortality.

Our difference-in-differences results from estimating equation (3), reported in Table 3, are consistent with these descriptive patterns. In columns 1 to 3, the dependent variable is the number of infant deaths, and in columns 4 to 6, it is the infant mortality rate per 1,000 live births. Each coefficient in panels A to C is estimated by Poisson pseudo-maximum likelihood and comes from a separate regression that includes county-of-birth and year-of-birth fixed effects. Columns 1 and 4 present baseline two-way fixed effects models without weights or controls, columns 2 and

5 weight observations by the number of county-by-year births, and columns 3 and 6 include weights and the control variables described in Section III. Panel A presents pooled estimates, while panels B and C estimate effects separately for Black and White infants, respectively.²³ In the last row, we present *p*-values for the difference in coefficients presented in panels B and C that are estimated using a model that is fully interacted with race.

We find that exposure to Duke Endowment support around the time of birth reduced infant mortality by 6.1 percent to 8.1 percent (panel A). The effects are larger for Black infants compared with White infants. In our preferred specification (column 6), the estimated coefficients suggest that Duke support led to a 7.5 percent reduction in overall infant mortality, a 13.6 percent reduction in Black infant mortality, and a 4.7 percent reduction in White infant mortality (*p*-value: 0.11).²⁴ The difference between the race-specific estimates is statistically significant (with a *p*-value of 0.02). The effects on pooled infant mortality are similar in size across project types: new hospitals, additions to hospitals, the purchase of equipment, and the purchase of existing facilities—typically proprietary hospitals slated for conversion to a nonprofit (see online Appendix Figure E2). While we find large effects for mortality on the day of birth, capturing the impact of health care provided before, during, and immediately after delivery, we also find effects on infant mortality from day 1 to day 365 infant mortality. Thus, our results are not fully explained by health care provided at birth, suggesting that support from the Duke Endowment likely improved neonatal health care as well (see online Appendix Table G4).²⁵

The average treatment effects documented in Table 3 are further validated by event studies based on estimating equation (4). These results are presented in the bottom row of Figure 4. In each case, before any funding was allocated to a county, there are no significant differences in infant mortality between treatment and control groups, supporting our assumption of parallel trends. In contrast, in the posttreatment periods we do see marked declines in mortality starting from the second year after the initial appropriation.²⁶

²³ There are fewer observations in panels B and C because in five counties we do not observe Black births in all county-by-year cells. We drop these counties from the analysis split by race but include them in the pooled analysis in panel A. Pooled results are almost identical when run on the sample considered in panels B and C. We discuss the adjustments to birth or death counts in greater detail in online Appendix H.

²⁴ The estimates from our preferred specification are among the most conservative across a range of models that we consider (see Figure 5). In many of these specifications, and across alternatives reported in online Appendix D, we find larger and more precisely estimated effects, including statistically significant effects for White infant mortality. See Section VI for further discussion.

²⁵ Our results are also insensitive to excluding deaths in the first week or in the first month of infancy. As we follow published infant mortality statistics in restricting attention to live births, we show in online Appendix G that potential measurement error in recording stillbirths does not drive our results for mortality on the day of birth.

²⁶ In online Appendix G we further document effects on fertility measured by the number of births per 1,000 women (online Appendix Table G3) and on maternal mortality (online Appendix Table G2). Point estimates for fertility neither imply major changes in childbearing due to improved health care infrastructure nor suggest a cohort-composition changes concern that one might have when interpreting our main specifications that use the number of births to create both the infant mortality rate and to weight observations. In any case, our specifications with death counts as an outcome are unaffected by concerns related to the number of births. On the other hand, in online Appendix Table G2 we find evidence that maternal mortality declined by roughly 20 percent following a Duke capital appropriation. Unlike our infant mortality data, these data are not constructed from death certificates, but rather they are reported at the county-year level in North Carolina annual vital statistics reports. The data are only reported beginning in 1932, and thus, the sample includes roughly half the number of observations as in the main specification. Due to the limited number of observations and lack of microdata, we cannot conduct the same

Magnitudes.—As context for our estimates, we consider the overall changes in infant mortality over the study period. The pooled infant mortality rate in North Carolina declined by 35.9 percent (from 72.6 to 46.6 deaths per 1,000 live births) from the time period just before Duke’s intervention (1922–1926) to the end of the study period (1938–1942). Similar reductions of 37.9 percent and 34.8 percent were observed for Black and White infant mortality, respectively. Our treatment effect estimates imply that 21 percent of the reduction in pooled infant mortality could be attributed to Duke support. These back-of-the-envelope calculations differ considerably by race, as Duke support contributed to 35.9 percent of the reduction in Black infant mortality but only 13.6 percent of the reduction for White infants.

We can also compare our effect sizes to other estimates in the literature, especially those examining a variety of health-related interventions from the United States in the first half of the twentieth century. For example, the Sheppard-Towner Act reduced infant mortality by up to 1.9 deaths per 1,000 births, explaining 21 percent of the infant mortality rate decline from 1924 to 1929, with much of the aggregate effect driven by decreases in mortality among Black infants (Moehling and Thomasson 2014). On the other hand, midwifery licensing reforms had a modest effect on average infant mortality and no detectable relationship with White infant mortality. For nonwhite infants, the coefficients were larger at 9.6 percent and statistically significant (Anderson et al. 2020).

Our estimated effects are consistent in size with those observed in more recent health care interventions. For example, Goodman-Bacon (2018) examines the impact of Medicaid introduction in the 1960s and finds that a 1 percentage point change in the initial Aid to Families with Dependent Children (AFDC) rate in a state resulted in a 1.4 percent reduction in infant mortality, implying an ATT of about 30 percent for nonwhite infants. Similarly, a 30 percentage point increase in Medicaid eligibility for pregnant women between 1979 and 1992 decreased infant mortality by 8.5 percent (Currie and Gruber 1996).

Intensive Margin.—Using data on the exact amount of money appropriated and disbursed by the Duke Endowment, we also investigate the intensive-margin effects of financial support on infant mortality and conduct a cost-benefit analysis. We reestimate equation (3) by substituting cumulative appropriations or cumulative payments at the county-by-year level in place of a dummy treatment variable. The results of this analysis are presented in Table 4. Columns 1 and 2 show effects per US\$(2017)1 million of appropriations and payments, respectively, while column 3 shows the conversion rate between a dollar appropriated (i.e., a commitment to pay) and a dollar paid. We find that almost 90 percent of the promised appropriations were disbursed. When we consider the actual money spent by the foundation, we find reductions in the infant mortality rate in the range of 7.6 percent (White infants) to 8.9 percent (Black infants) per \$1 million.

Based on the amount appropriated and spent, one life was saved for every \$19,110 appropriated or \$17,092 spent (in 2017 dollars), implying that the Endowment’s intervention passes any cost-benefit analysis using a reasonable estimate for a

battery of robustness tests that we do for infant mortality. We view these results as supportive evidence that Duke funding improved health for individuals other than infants.

TABLE 4—EFFECT OF DUKE SUPPORT ON INFANT MORTALITY RATE: INTENSIVE-MARGIN ESTIMATES

	Y_{ct}^R = Infant mortality rate		Y_{ct}^R = Cumulative payments
	Appropriations (1)	Payments (2)	Appropriations (3)
<i>Panel A. Pooled infant mortality rate</i>			
Percent effect from \$1 million of Duke support	−7.16 (0.97)	−7.71 (1.18)	
Cumulative appropriations			0.87 (0.04)
Observations	2,100	2,100	2,100
<i>Panel B. Black infant mortality rate</i>			
Percent effect from \$1 million of Duke support	−9.05 (1.66)	−8.91 (1.56)	
Cumulative appropriations			0.87 (0.03)
Observations	1,995	1,995	1,995
<i>Panel C. White infant mortality rate</i>			
Percent effect from \$1 million of Duke support	−6.51 (1.20)	−7.64 (1.63)	
Cumulative appropriations			0.86 (0.04)
Observations	1,995	1,995	1,995
<i>p</i> -value for difference by race	0.34	0.72	0.35
County of birth fixed effects	Yes	Yes	Yes
Year of birth fixed effects	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Weights	Yes	Yes	Yes

Notes: Estimates from econometric specifications based on column 6 of Table 3. The first two columns test for an intensive-margin effect, with the independent variable of interest being cumulative appropriations or cumulative payments to hospitals rather than a binary variable for ever having had an appropriation from the Duke Endowment. The dependent variable in columns 1 and 2 is the infant mortality rate per 1,000 live births. The third column looks at the relationship between cumulative appropriations and cumulative payments. Panels B and C drop counties that ever have zero race-specific births between 1922 and 1942. The bottom row of panel C presents a *p*-value from the interaction of race with our treatment variable from a model that fully interacts all variables with race. Standard errors are estimated using the delta method and are clustered at the county level.

VSL.²⁷ Most other papers in the literature lack the detailed cost data needed to provide similar estimates, with the notable exceptions of Currie and Gruber (1996) and Goodman-Bacon (2018), who find that an infant death could be averted at a cost of about \$1.8 million and \$171,000 (in 2017 dollars), respectively, suggesting that the investment in hospital infrastructure during the 1920s to 1940s might have been more cost-effective than Medicaid's introduction in the 1960s or its later expansions in the 1980s and 1990s.

D. Interaction with Medical Innovation

We now ask if the effects of Duke support were magnified (complements) or diminished (substitutes) by medical innovation in the form of sulfa drugs—the first modern treatment that could effectively control and fight bacterial infections. To the

²⁷For example, Ashenfelter and Greenstone (2004) estimate VSL at \$2.8 million, while in a review article Viscusi (2018) finds a median VSL in the United States at \$10.3 million.

extent that sulfa drugs reduce mortality (Jayachandran, Lleras-Muney, and Smith 2010), we would expect the two inputs to be complements. On the other hand, easy access to effective outpatient treatments such as antibiotics might lead to a shift in health care provision, rendering hospital access—the primary goal of Duke support—less important, thus leading to the two inputs being substitutes.

We first estimate a shift-share model of sulfa drugs on infant mortality. Since pneumonia was highly treatable with sulfa, we use baseline pneumonia mortality as the share factor. Specifically, the variable *Pneumonia mortality*_c is the average county-level pneumonia mortality for the years 1922 to 1926. The shift factor is an indicator variable *Post sulfa*_t that takes the value of 1 for the years 1937 and beyond when sulfa drugs were available. Results from this analysis suggest that sulfa drugs reduced pooled infant mortality by 5.2 percent (see online Appendix Table G6). To assess whether the two inputs—Duke support and sulfa drugs—are complements, substitutes, or independent treatments, we interact our rollout (Duke support) and shift-share (sulfa drugs) difference-in-differences designs and estimate the following equation:

$$\begin{aligned}
 (5) \quad Y_{ct}^R = & \exp[\gamma_0 + \gamma_1 \mathbf{1}\{t \geq \textit{First appropriation year}\}_{ct} \\
 & + \gamma_2 \mathbf{1}\{t \geq \textit{First appropriation year}\}_{ct} \times \textit{Post sulfa}_t \\
 & + \gamma_3 \mathbf{1}\{t \geq \textit{First appropriation year}\}_{ct} \\
 & \times \textit{Pneumonia mortality}_c + \gamma_4 \textit{Pneumonia mortality}_c \\
 & \times \textit{Post sulfa}_t + \gamma_5 \mathbf{1}\{t \geq \textit{First appropriation year}\}_{ct} \\
 & \times \textit{Pneumonia mortality}_c \times \textit{Post sulfa}_t + \zeta_c + \eta_t + \Theta \mathbf{X}_{ct}] \varepsilon_{ct},
 \end{aligned}$$

where Y_{ct}^R , ζ_c , η_t , $\mathbf{X}_{ct}$, and $\mathbf{1}\{t \geq \textit{First appropriation year}\}_{ct}$ are defined as in equation (3), while *Pneumonia mortality*_c and *Post sulfa*_t are defined as described above. The main parameter of interest in this equation is γ_5 , which describes how the effects of receiving Duke funding change after sulfa drugs become available, accounting for variation in baseline pneumonia mortality across counties. The auxiliary parameters γ_1 to γ_4 allow us to compute Duke funding effects separately for the pre- and post-sulfa periods (shift factor) and separately for counties with high and low pneumonia mortality (share factor). We cluster the standard errors (ε_{ct}) by county of birth to account for correlated errors within a county.

Table 5 reports linear combinations of these coefficients for ease of interpretation. Panel A presents an estimate of the interaction between Duke funding and the introduction of sulfa drugs, comparing counties at the twenty-fifth percentile of baseline pneumonia mortality to those at the seventy-fifth percentile. Results indicate that gains from Duke support increased with the availability of medical innovation and did so to a greater extent in counties with high baseline pneumonia mortality, implying that these two inputs are complements. Our results are in line with Alsan and Goldin (2019), who find compounding effects of clean water and sewerage infrastructure.

TABLE 5—INTERACTION OF ROLLOUT OF DUKE SUPPORT AND DISCOVERY OF SULFA DRUGS

	$Y_{ct}^R = \text{Infant mortality rate}$		
	Pooled (1)	Black (2)	White (3)
<i>Panel A. Interaction of Duke rollout $\times$ sulfa shift-share DiD</i>			
Post-pre sulfa <i>Pneumonia</i> _{0.75} –	–15.09	–12.57	–16.20
Post-pre sulfa <i>Pneumonia</i> _{0.25} (γ_5)	(4.94)	(6.54)	(5.32)
<i>Panel B. Duke versus no Duke</i>			
Post-sulfa, <i>Pneumonia</i> _{0.75}	–15.77	–19.44	–14.82
$\gamma_1 + \gamma_2 + \eta_{0.75} \times (\gamma_3 + \gamma_5)$	(3.96)	(5.58)	(4.07)
Post-sulfa, <i>Pneumonia</i> _{0.25}	1.51	–4.07	2.47
$\gamma_1 + \gamma_2 + \eta_{0.25} \times (\gamma_3 + \gamma_5)$	(7.05)	(9.29)	(7.43)
Pre-sulfa, <i>Pneumonia</i> _{0.75}	–5.57	–11.60	–2.72
$\gamma_1 + \eta_{0.75} \times \gamma_3$	(2.37)	(2.75)	(3.34)
Pre-sulfa, <i>Pneumonia</i> _{0.25}	–3.37	–7.96	–1.93
$\gamma_1 + \eta_{0.25} \times \gamma_3$	(3.29)	(4.81)	(4.15)
<i>Panel C. (Duke versus no Duke) $\times$ (Pre versus post sulfa) or (Duke versus no Duke) $\times$ (Pneumonia IQR)</i>			
<i>Pneumonia</i> _{0.75} , Post-pre sulfa	–10.81	–8.87	–12.44
$\gamma_2 + \eta_{0.75} \times \gamma_5$	(4.18)	(5.52)	(4.51)
<i>Pneumonia</i> _{0.25} , Post-pre sulfa	5.05	4.23	4.49
$\gamma_2 + \eta_{0.25} \times \gamma_5$	(7.07)	(7.90)	(8.20)
Post-sulfa, <i>Pneumonia</i> _{IQR}	–17.02	–16.02	–16.88
$(\eta_{0.75} - \eta_{0.25}) \times (\gamma_3 + \gamma_5)$	(5.77)	(7.76)	(5.63)
Pre-sulfa, <i>Pneumonia</i> _{IQR}	–2.27	–3.95	–0.81
$(\eta_{0.75} - \eta_{0.25}) \times \gamma_3$	(2.88)	(3.59)	(3.08)
Observations	2,100	1,995	2,100

Notes: Estimates in each column come from separate Poisson pseudo-maximum likelihood regressions. Each panel reports exponentiated linear combinations of coefficients from equation (5). Because we report exponentiated linear combinations of the underlying coefficients, linear combinations of this nonlinear transform may not be equal to the reported coefficient due to Jensen's inequality (i.e., $\exp(a + b) \leq \exp(a) + \exp(b)$). Column 2 drops the five counties that ever have zero Black births between 1922 and 1942. The dependent variable is the number of infant deaths per 1,000 live births in county c and year t birth cohort. Duke exposure is a binary variable that takes the value of 1 in all years starting in the first year that a county received a capital appropriation from the Endowment. Sulfa exposure (shift) is measured by an indicator that equals to 1 for all years from 1937 onward. The share factor is the average pneumonia mortality per 100,000 individuals in a county between 1922 and 1926. *Pneumonia*_{IQR} $(\eta_{0.75} - \eta_{0.25})$ denotes the difference between the seventy-fifth and twenty-fifth percentiles of baseline pneumonia mortality. All regressions include county and year fixed effects. Control variables include percent illiterate, percent Black, percent other race, percent urban, retail sales per capita, and presence of a county health department. Observations are weighted by the number of births in a county and year. Standard errors are estimated using the delta method and are clustered at the county level.

In panels B and C, we break down the interaction coefficient to examine the effects of Duke support separately in areas with high baseline pneumonia mortality and those without, as well as before and after the introduction of sulfa drugs.

We observe no statistically significant effects of Duke funding in counties with low baseline pneumonia mortality, either before or after the introduction of sulfa drugs (rows 2 and 4 in panel B). Furthermore, the difference in the Duke effect between the pre- and post-sulfa periods in these counties is small and statistically insignificant (row 2 of panel C).

However, in counties with high baseline pneumonia mortality, most of the estimated coefficients are statistically significant both before and after sulfa drugs

become available (rows 1 and 3 of panel B). Moreover, these estimated coefficients are larger in the postperiod. This reflects the complementary nature of hospital modernization with medical innovation, which in this case may be explained by the fact that the Duke Endowment attracted highly qualified physicians to the area who were better able to leverage the new technology.

V. Long-Run Mortality

Given the broad literature suggesting that early-life interventions can have long-run impacts, we next examine if the health gains from Duke support translated into reductions in long-run mortality for those exposed to funding around the time of their birth. On one hand, if reductions in early-life mortality are positively associated with health capital, and by extension human capital and income, then we would also expect to find a decrease in mortality at older ages. On the other hand, if vulnerable individuals induced by Duke funding to survive past the first year of life likewise survived past age 55, then we would expect to find null effects or even increases in long-run mortality.²⁸

We implement the same difference-in-differences framework from Section IV based on a comparison of cohorts exposed to Duke financial support around the time of birth to those without such support to estimate effects on long-run mortality using the following equation:

$$(6) \quad Y_{ctf}^R = \exp[\beta_0 + \beta_1 \mathbf{1}\{t \geq \text{First appropriation year}\}_{ct} + \zeta_c \cdot \nu_a + \eta_t \cdot \nu_a + \Theta \mathbf{X}_{ctf}] \varepsilon_{ctf},$$

which is similar to equation (3) except (i) we stack birth-county-by-year observations over follow-up ages f and (ii) due to the incompleteness of the Numident prior to 1988 (as described in Section II), we rely on the 1932 to 1941 birth cohorts, which can all be observed for nine follow-up years between ages 56 and 64. Y_{ctf}^R is the number of deaths for those born in county c and in birth cohort t and observed in follow-up year f . All specifications include birth county by follow-up age ($\zeta_c \cdot \nu_a$) and birth year by follow-up age ($\eta_t \cdot \nu_a$) fixed effects, along with a vector of controls, $\mathbf{X}_{ctf}$.²⁹ We cluster standard errors at the birth county level and weight each observation by the number of births in each birth-county-by-year cohort.

We present the long-run results in Table 6, structured similarly to Table 3 but with the number of later-life deaths as the dependent variable.³⁰ We present pooled estimates in panel A, with separate effects for Black and White adults shown in panels B and C, respectively. These coefficients represent the impact of treatment around the time of birth on mortality between ages 56 and 64, conditional on survival to

²⁸ All our long-run analyses are conditional on survival to at least age 56 for reasons described in Section II.

²⁹ Since we observe each county-by-year-of-birth cohort at ages 56 to 64 during follow-up, we allow baseline controls to flexibly vary with follow-up age. These additional control variables are included in the vector $\mathbf{X}_{ctf}$.

³⁰ We prefer to use death counts in this analysis since we do not have data on the surviving number of people from each county-year of birth cohort, which would be ideal for computing death rates. In online Appendix Table F1, we find that results are substantively unchanged when we use the death rate per 1,000 live births in each county-by-year of birth cohort as the outcome.

TABLE 6—EFFECT OF DUKE SUPPORT AROUND TIME OF BIRTH ON LONG-RUN MORTALITY AT AGES 56 TO 64

	$Y_{ct}^R = \text{Long-run deaths}$		
	(1)	(2)	(3)
<i>Panel A. Pooled long-run deaths</i>			
Percent effect from Duke (= 1)	−7.66 (3.08)	−10.07 (2.66)	−8.99 (2.81)
Observations	9,000	9,000	9,000
<i>Panel B. Black long-run deaths</i>			
Percent effect from Duke (= 1)	−8.04 (3.57)	−8.77 (2.80)	−7.58 (3.54)
Observations	8,150	8,150	8,150
<i>Panel C. White long-run deaths</i>			
Percent effect from Duke (= 1)	−6.93 (3.83)	−10.56 (2.92)	−9.53 (2.86)
Observations	8,630	8,630	8,630
<i>p</i> -value for difference by race	0.82	0.61	0.63
County of birth × Age FE	Yes	Yes	Yes
Year of birth × Age FE	Yes	Yes	Yes
Weights	No	Yes	Yes
Controls	No	No	Yes

Notes: Each coefficient comes from a separate Poisson pseudo-maximum likelihood regression. The unit of observation is a birth county by birth year by follow-up age triplet. Birth cohorts are restricted to 1932 to 1941. Deaths are restricted to ages 56 to 64 and years 1988 to 2005. The dependent variable is the number of age-specific deaths. Each coefficient represents the percent reduction in long-run mortality due to receiving a capital appropriation from the Duke Endowment around the time of birth. A coefficient of −10 would mean that long-run mortality declines by 10 percent. Coefficients reported in the table are transformed in the following way: $100 \times [\exp(\beta) - 1]$. Control variables in column 3 include flexible interactions of age fixed effects with percent illiterate, percent Black, percent other race, percent urban, retail sales per capita, and presence of a county health department. In columns 2 to 3, the weights are county-by-year birth cohort size. All specifications include county-of-birth fixed effects interacted with follow-up age fixed effects and year-of-birth fixed effects interacted with follow-up age fixed effects. Panels B and C drop counties that ever have zero race-specific births between 1932 and 1941. Observations differ across the two samples, as observations are dropped that are perfectly separated by either county-of-birth by follow-up age fixed effects or year-of-birth by follow-up age fixed effects. The bottom row of panel C presents a *p*-value from the interaction of race with our treatment variable from a model that fully interacts all variables with race. Standard errors are estimated using the delta method and are clustered at the county level.

age 56. We find consistently negative and statistically significant coefficients. In our preferred specification (column 3), the average treatment effect is a 9.0 percent reduction in long-run mortality. This effect size is slightly larger than our preferred estimate from Table 3, but it is sufficiently close for us to conclude that many of the health benefits gained during infancy appear to have persisted into adulthood. We present and discuss event study estimates corresponding to these results in Section F.1 of the online Appendix.

The long-run effects on Black and White mortality are similar across specifications, in contrast to the larger effects observed on Black infant mortality in Section IV. This difference may arise from the fact that some of the most vulnerable Black infants, whose survival beyond the first year of life could have been attributed to the intervention, did not live long enough to be included in our long-run sample, which is consistent with known disparities in life expectancy at birth between Black

and White babies.³¹ Alternatively, the marginal effects of Duke support may simply not differ by race beyond age 55.

Our long-run results by race differ from Goodman-Bacon (2021b), who observed that early-age Medicaid eligibility reduced non-AIDS adult mortality before age 37 by 14.5 percent for White individuals and 8.7 percent for nonwhite individuals.³² However, our pooled long-run estimates are similar in magnitude to the long-run impact of access to in-hospital deliveries (Fischer, Karlsson, and Prodromidis 2021) and much larger than the effects of a nurse home visiting program (Hjort, Sølvesten, and Wüst 2017). We conclude that Duke support improved the long-run welfare of those treated around the time of birth. Yet we are almost certainly understating the population-wide impact of Duke support on long-run mortality since it may have also benefited groups who only had access to it *after* infancy, and Duke access may have improved health at later ages for those who already had access to Duke-supported hospitals at birth.

Permanence of Effect.—While we find evidence of a long-term reduction in mortality for individuals treated at birth, a question remains: how enduring were the effects of Duke support on infant mortality and hospital capacity? In terms of potential outcomes, it is possible that treated areas could have eventually reached parity with untreated areas in the absence of the treatment, implying temporary effects of Duke support. Section L of the online Appendix explores this issue by extending the sample until 1950 for the hospital sector analysis and 1962 for the infant mortality analysis. Our findings provide clear evidence that the effects of Duke support are persistent and do not diminish over time. Consequently, our results should be interpreted as a permanent change in infant mortality and hospital capacity between treated and untreated counties.

VI. Robustness Checks

Point estimates and statistical significance are robust to the choice of estimator, specification, transformation of the dependent variable, choice of the control group, and other sample restrictions, such as dropping small counties or changing the endpoints of the sample.³³ We provide a visual overview of many of these checks for our infant mortality results in Figure 5.

Robust Estimators.—Our event studies in Figures 2 to 4 are insensitive to the use of stacked regression (Cengiz et al. 2019), extended two-way fixed effects

³¹For example, in 1900, Black and White life expectancy at birth was 33 and 48 years, while by 2017, it increased to 61 and 69 years for these two groups, respectively (National Center for Health Statistics 2017). Data for 1900 to 1950 is more scarce, but Ewbank (1987) suggests that near the end of our sample period in 1939 to 1940, life expectancy at birth in North Carolina for Black and White infants was 54 and 64 years, respectively.

³²Interestingly, using the same variation, Goodman-Bacon (2018) finds reductions in infant and child mortality only for nonwhite children.

³³When treatment is staggered, DiD estimates place greater weight on treatment groups that are treated closer to the middle of the panel and may be sensitive to panel length. Even though over half of the Duke rollout occurred within the first two years of program launch, the extended rollout period could potentially bias our estimates due to dynamic treatment effects that vary across treatment timing groups (Goodman-Bacon 2021a; Baker, Larker, and Wang 2022).

(Wooldridge 2021), and the Callaway and Sant’Anna (2021) estimators that are designed to address concerns about treatment effect heterogeneity and bias in a difference-in-differences setting with staggered treatment adoption. The results are similar across all of the estimators, and, as before, we do not see evidence of pre-trends in our preferred specifications (see online Appendix Tables B1, C1, and D1 for static difference-in-differences estimates).

The validity of our event studies is further supported by the Goodman-Bacon (2021a) decomposition, which reveals that 2x2 DiD comparisons of treated versus untreated units receive the lion’s share of the weight in the overall two-way fixed effects DiD estimate (see online Appendix Table I1 and Figure I1). Our main infant mortality results are also invariant to specifying the dependent variable in logs (online Appendix Table D2), while stacked Poisson event studies are insensitive to adjusting the width of the event-time window from ± 3 to ± 6 and including control variables (online Appendix Figure D1), providing support for parallel pre-trends.

The effects on White infant mortality are comparable to or larger than those estimated using our preferred specification (Table 3), ranging from -4.3 percent to -18.4 percent (online Appendix Table D1), and are generally statistically significant, which suggests that our preferred estimates may be especially conservative. At the same time, estimates for Black infants are always statistically significant and almost universally much larger than estimates for White infants.

Using Counties in Other Southern States Ineligible for Duke Funding as Controls.—We address concerns about selection into treatment and spillover effects with a variety of analyses that add other non-Carolina Southern counties. For example, North Carolina counties in the control group may have been ill prepared to take advantage of Duke funding or simply may not have had any not-for-profit hospitals that could be supported. In addition, higher-quality physicians could have moved from control to treated counties, or the state government could have redirected resources from counties that did not get Duke support to those that did (or vice versa). In the presence of such effects, our difference-in-differences estimates may overstate the benefits of the intervention. Thus, we define alternate sets of control group counties that should not have been affected by any spillovers or selection.

Up to the 1930s, infant mortality rates were relatively similar across ever-treated North Carolina counties and other Southern counties that were ineligible for Duke funding because they were outside of the Carolinas (see online Appendix Figure J1). However, as Duke support expanded over the remainder of the decade, North Carolina experienced a large decline in infant mortality for both White and Black infants that was not observed in other Southern states. Our results are robust to alternative DiD (online Appendix Table F3) and event study (online Appendix Figure J2) specifications that use these non-Carolina Southern counties as controls, although these estimates suggest much more similar effects by race.

We also use propensity score matching to construct an alternative set of control counties from the Southern states and to conduct a placebo analysis. First, we only select control counties from other Southern states that are observably similar to treated North Carolina counties. Naturally, these control counties are ineligible for Duke founding. These results are presented in online Appendix Tables M2 to M4 and are similar to those in online Appendix Table F3. Second, our placebo exercise

uses only non-Carolina Southern counties that we divide into those that look the most and the least like our true treated counties. We then assign treatment to the former group and estimate our preferred infant mortality regressions. As expected, results presented in online Appendix Table M1 yield small, inconsistently signed, and almost never statistically significant coefficients.

Instrumental Variables.—As another layer of evidence showing that our findings are not driven by selection, we consider a version of our intensive-margin treatment effects in an instrumental variables setting. The instrument interacts two exogenous sources of variation. The first source is temporal and leverages the fact that as income from the Endowment's investments increased, so did the ability of the Endowment to appropriate and spend funds. The second source is cross-sectional and exploits the fact that North Carolina counties with a not-for-profit hospital at the time the Endowment was founded had an advantage over other counties because they were eligible for capital appropriations to improve existing hospitals.

Compared to our preferred intensive-margin results, the instrumental variables framework yields larger point estimates that are statistically significant at conventional levels (see online Appendix Table J1). We report both Anderson and Rubin (1949) 95 percent confidence sets and tF 95 percent confidence intervals (Lee et al. 2022) for each IV estimate, demonstrating that the strength of our first stage is not an issue. Results suggest that for every \$1 million that the Duke Endowment appropriated in a county, the infant mortality rate declined between 14.6 and 17.4 percent. For payments, the effect sizes are between 16.2 and 17.4 percent per \$1 million. If selection into treatment, targeting of funds, or trends in counties with not-for-profit hospitals were driving our results, we would expect the instrumental variable results to attenuate these treatment effect estimates (see online Appendix J for additional details). The finding of a larger, negative effect mitigates these concerns and provides additional evidence in favor of our overall argument.

Randomization Inference.—Next, we explore whether the nonrandom selection of counties that received Duke support or the nonrandom timing of treatment may have distorted our main estimates for infant mortality. In online Appendix Figure K1, we present the results from a randomization inference exercise based on the specification from column 5 of Table 3). In this procedure, we randomly assign 48 North Carolina counties to the treatment path of the 48 counties that actually received treatment. As expected under the assumption of no selection bias, the average placebo estimate is symmetrical and centered around zero.

We also use this exercise to construct two-sided randomization inference based *p*-values for our preferred point estimates.³⁴ Our results remain statistically significant, indicating that they are not driven by spurious time trends in infant mortality across all counties or by spillovers across treated versus untreated counties in North Carolina.

³⁴We do so by comparing the absolute value of each treatment effect estimate to the absolute value of pseudo-treatment effect estimates drawn from 10,000 different randomly drawn treatment assignments. The *p*-value is the percent of random draws that are larger in absolute value than the estimated treatment effect.

Long-Run Mortality.—Our long-run mortality results are unaffected if we collapse the data to the cohort level and use cumulative mortality at ages 56 to 64 as the outcome rather than including separate observations for mortality at each follow-up age (online Appendix Table F2). They are robust to including non-Carolina counties that were ineligible for Duke funding as controls (online Appendix Table F3) and adding South Carolina to the treatment group (online Appendix Table F4).

VII. Conclusions

The barriers that limit access to effective health care in the United States today—including a lack of health care providers and facilities in rural and impoverished areas, financial constraints, unequal access based on race and socioeconomic status, inadequate insurance coverage, and structural obstacles such as transportation difficulties—are not unique to the modern era. These challenges were equally prevalent almost 100 years ago in 1924, when James Buchanan Duke established the Duke Endowment with the intention of alleviating such burdens in the Carolinas.

In this paper, we leverage exogenous variation in support provided by the foundation—including the building of new hospitals, expansions and upgrades of existing facilities, modernization of equipment, and improved managerial practices—to study its consequences on infant and long-run mortality, with a particular focus on understanding racial disparities in health outcomes.

Our findings reveal that the Duke Endowment's support had a substantial impact on the health care infrastructure of North Carolina. It led to an expansion in the number of not-for-profit hospitals and an increase in available beds. Furthermore, it significantly increased the number of practicing physicians in treated counties, effectively replacing less-qualified doctors with highly trained counterparts. Furthermore, we show that counties supported by Duke disproportionately benefited from the invention of sulfa drugs, providing some of the first causal evidence that hospital funding and medical innovation are complements rather than substitutes.

As a result of health care infrastructure improvements facilitated by the Duke Endowment's support, infant mortality decreased by 7.5 percent. This impact was more pronounced for Black infants, who experienced a 13.6 percent reduction in mortality compared to a 4.7 percent decrease for White infants. These gains for Black infants succeeded in narrowing the pre-intervention racial gap in infant mortality by one-quarter. Furthermore, we provide evidence that the health benefits of support from the Duke Endowment persisted until at least modern retirement age for those exposed around the time of birth, reducing the average mortality between the ages 56 and 64 by 9.0 percent.

We calculate that each infant life was saved at a remarkably low cost of \$US(2017)17,092. However, it is important to note that in today's context, where baseline health and health care access are considerably higher, a similar intervention would likely yield a smaller impact at a higher cost. Nevertheless, given the ongoing challenges in the provision of health care to rural areas, these results highlight the potential for investments in hospitals to still yield significant and cost-effective impacts.

Although the intervention we study occurred in the first half of the twentieth century in the United States, we view our findings as not only contributing to expanding

basic knowledge about the development of US health care but also to historical and contemporaneous public policy. Despite the viewpoint that modern medicine did not materially affect health outcomes in the United States until the 1950s, our results are *proof of concept* that investments in for-the-times modern hospital infrastructure were indeed productive in both the short and long run. Furthermore, what was considered a modern hospital in the United States during the 1930s and 1940s pales in comparison to the currently available technology, but it may not be that dissimilar to what is offered in less developed nations and especially in underserved and rural regions of these countries. To the extent that our findings generalize to such settings, they may guide policymakers in improving infant and adult health in a cost-effective way.

The Duke Endowment's financial support for health care in the Carolinas also *served as both inspiration and model* for the Hill-Burton Act of 1946 (Durdin 1998), which was a federal program that subsidized large-scale construction of hospitals in underserved locations throughout the United States to ensure that each state had an average of 4.5 hospital beds per 1,000 residents. Over the life of the program, the federal government spent over \$3.7 billion on that goal (Chung, Gaynor, and Richards-Shubik 2017), but very little is known about its effects on health or human capital outcomes of individuals that it touched. While Duke support was the exception rather than the norm during the first half of the twentieth century, and thus likely could not have contributed to the aggregate trends in mortality, it appears plausible that a nontrivial fraction of the massive decline in infant mortality in the United States between 1945 and 1975 (from 44 to 18 deaths per 1,000 live births) might be attributable to this intervention given that the Hill-Burton program was modeled on the Duke Endowment. We view this as both a policy-relevant and promising agenda for future research, which could also inform questions about the scale-up of smaller *trial programs* to the national level.

Finally, it is important to point out that we focus on a very specific research question and conduct analysis that is limited in part by data availability. Since we find large and positive health effects that persist in the long run across the Carolinas, it is possible that other economic outcomes of the treated individuals, such as human capital or wages, also improved. Furthermore, since the support from the Duke Endowment was not targeted at pregnant women or infants and because it induced permanent changes in access to modern hospitals in supported communities, it plausibly benefited other groups beyond those considered in this paper. This means that our estimates are likely to be a conservative lower bound for the total welfare improvements. We view the questions about the cumulative effects of the program, effects on other outcomes, and impacts on other age groups as important future extensions of this work.

REFERENCES

- Aaronson, Daniel, Fabian Lange, and Bhashkar Mazumder. 2014. "Fertility Transitions Along the Extensive and Intensive Margins." *American Economic Review* 104 (11): 3701–24.
- Aaronson, Daniel, and Bhashkar Mazumder. 2011. "The Effect of Rosenwald Schools on Black Achievement." *Journal of Political Economy* 119 (5): 821–88.
- Almond, Douglas, Janet Currie, and Valentina Duque. 2018. "Childhood Circumstances and Adult Outcomes: Act II." *Journal of Economic Literature* 56 (4): 1360–1446.
- Alsan, Marcella, and Claudia Goldin. 2019. "Watersheds in Child Mortality: The Role of Effective Water and Sewerage Infrastructure, 1880–1920." *Journal of Political Economy* 127 (2): 586–638.

- American Medical Association.** 1942a. "Hospital Service in the United States, 1920-1942" in *Journal of the American Medical Association*, Chicago, IL: American Medical Association.
- American Medical Association.** 1942b. *American medical directory, 1906-1942*, Chicago, IL: American Medical Association.
- Anderson, D. Mark, Ryan Brown, Kerwin Kofi Charles, and Daniel I. Rees.** 2020. "Occupational Licensing and Maternal Health: Evidence from Early Midwifery Laws." *Journal of Political Economy* 128 (11): 4337-83.
- Anderson, D. Mark, Kerwin Kofi Charles, and Daniel I. Rees.** 2021. "The Federal Effort to Desegregate Southern Hospitals and the Black-White Infant Mortality Gap." NBER Working Paper 27970.
- Anderson, D. Mark, Kerwin Kofi Charles, and Daniel I. Rees.** 2022. "Reexamining the Contribution of Public Health Efforts to the Decline in Urban Mortality." *American Economic Journal: Applied Economics* 14 (2): 126-57.
- Anderson, D. Mark, Kerwin Kofi Charles, Daniel I. Rees, and Tianyi Wang.** 2021. "Water Purification Efforts and the Black-white Infant Mortality Gap, 1906-1938." *Journal of Urban Economics* 122: 103329.
- Anderson, T. W., and Herman Rubin.** 1949. "Estimation of the Parameters of a Single Equation in a Complete System of Stochastic Equations." *Annals of Mathematical Statistics* 20 (1): 46-63.
- Arias, Elizabeth.** 2014. "United States life tables, 2010." *National Vital Statistics Reports* 63 (7): 1-63.
- Arias, Elizabeth, Betzaida Tejada-Vera, Farida Ahmad, and Kenneth D. Kochanek.** 2021. "Provisional Life Expectancy Estimates for 2020." *Vital Statistics Rapid Release* 15: 50-61.
- Arthi, Vellore, Brian Beach, and W. Walker Hanlon.** 2022. "Recessions, Mortality, and Migration Bias: Evidence from the Lancashire Cotton Famine." *American Economic Journal: Applied Economics* 14 (2): 228-55.
- Ashenfelter, Orley, and Michael Greenstone.** 2004. "Using Mandated Speed Limits to Measure the Value of a Statistical Life." *Journal of Political Economy* 112 (1): S226-67.
- Bailey, Martha, Karen Clay, Price Fishback, Michael Haines, Shawn Kantor, Edson Severnini, and Anna Wentz.** 2015. "U.S. County-Level Natality and Mortality Data, 1915-2007." ICPSR. <https://doi.org/10.3886/ICPSR36603.v2> (accessed June 10, 2021).
- Baker, Andrew., David F. Larker, and Charles C. Y. Wang.** 2022. "How Much Should We Trust Staggered Difference-in-Differences Estimates?" *Journal of Financial Economics* 144 (2): 370-95.
- Black, Dan A., Seth G. Sanders, Evan J. Taylor, and Lowell J. Taylor.** 2015. "The Impact of the Great Migration on Mortality of African Americans: Evidence from the Deep South." *American Economic Review* 105 (2): 477-503.
- Bleakley, Hoyt.** 2007. "Disease and Development: Evidence from Hookworm Eradication in the American South." *Quarterly Journal of Economics* 122 (1): 73-117.
- Bloom, Nicholas, and John Van Reenen.** 2007. "Measuring and Explaining Management Practices Across Firms and Countries." *Quarterly Journal of Economics* 122 (4): 1351-1408.
- Callaway, Brantly and Pedro H. C. Sant'Anna.** 2021. "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics* 225 (2): 200-230.
- Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer.** 2019. "The Effect of Minimum Wages on Low-Wage Jobs." *Quarterly Journal of Economics* 134 (3): 1405-54.
- Chen, Jiafeng, and Jonathan Roth.** 2023. "Logs with Zeros? Some Problems and Solutions." Unpublished.
- Chung, Andrea Park, Martin Gaynor, and Seth Richards-Shubik.** 2017. "Subsidies and Structure: The Lasting Impact of the Hill-Burton Program on the Hospital Industry." *Review of Economics and Statistics* 99 (5): 926-43.
- CMS.** 2022. *NHE Fact Sheet* | CMS. Washington, DC: US Centers for Medicare & Medicaid Services.
- Collins, William J., and Melissa A. Thomasson.** 2004. "The Declining Contribution of Socioeconomic Disparities to the Racial Gap in Infant Mortality Rates, 1920-1970." *Southern Economic Journal* 70 (4): 746-76.
- Cook, Lisa D., Trevon D. Logan, and John M. Parman.** 2014. "Distinctively Black Names in the American Past." *Explorations in Economic History* 53: 64-82.
- Cook, Lisa D., Trevon D. Logan, and John M. Parman.** 2016. "The Mortality Consequences of Distinctively Black Names." *Explorations in Economic History* 59: 114-25.
- Correia, Sergio, Paulo Guimarães, and Tom Zylkin.** 2020. "Fast Poisson Estimation with High-Dimensional Fixed Effects." *Stata Journal* 20 (1): 95-115.
- Costa, Dora L.** 2015. "Health and the Economy in the United States from 1750 to the Present." *Journal of Economic Literature* 53 (3): 503-70.
- Craig, Lee.** 2006. "Consumer Expenditures." In *Historical Statistics of United States: Millennial Edition*, Vol. 3, edited by Susan B. Carter, Scott Sigmund Gartner, Michael R. Haines, Alan L. Olmstead, Richard Sutch, and Gavin Wright. Cambridge: Cambridge University Press.

- Currie, Janet, and Jonathan Gruber. 1996. "Saving Babies: The Efficacy and Cost of Recent Changes in the Medicaid Eligibility of Pregnant Women." *Journal of Political Economy* 104 (6): 1263–96.
- Cutler, David M. 2005. *Your Money or Your Life: Strong Medicine for America's Health Care System*. Oxford: Oxford University Press.
- Duke Endowment. 1925. *Annual Report of the Hospital Section of the Duke Endowment for the year 1925*. Charlotte, NC: The Duke Endowment.
- Duke Endowment. 1934. *Annual Report of the Hospital Section of the Duke Endowment for the year 1934*. Charlotte, NC: The Duke Endowment.
- Durbin, Robert F. 1998. *Lasting Legacy to the Carolinas*. Durham, NC: Duke University Press.
- Elder, Todd E., John H. Goddeeris, and Steven J. Haider. 2016. "Racial and Ethnic Infant Mortality Gaps and the Role of Socio-Economic Status." *Labour Economics* 43: 42–54.
- Esteves, Rui, Kris James Mitchener, Peter Nencka, and Melissa A. Thomasson. 2022. "Do Pandemics Change Healthcare? Evidence from the Great Influenza." NBER Working Paper 30643.
- Ewbank, Douglas. 1987. "History of Black Mortality and Health before 1940." *Milbank Quarterly* 65 (1): 100–128.
- Ferrell, J. Atkinson, and P. A. Mead. 1936. *History of County Health Organizations in the United States, 1908-33*. Washington, DC: US Government Printing Office.
- Figinski, Theodore F., and Erin Troland. 2020. "Health Insurance and Hospital Supply: Evidence from 1950s Coal Country." Unpublished.
- Fischer, Martin, Martin Karlsson, and Nikolaos Prodromidis. 2021. "The Long-term Effects of Hospital Deliveries." Unpublished.
- Fischer, Stephanie, Heather Royer, and Corey White. 2022. "Health Care Centralization: The Health Impacts of Obstetric Unit Closures in the US." NBER Working Paper 30141.
- Fishback, Price V., William C. Horrall, and Shawn Kantor. 2005. "Did New Deal Grant Programs Stimulate Local Economies? A Study of Federal Grants and Retail Sales During the Great Depression." *Journal of Economic History* 65 (1): 36–71.
- Gamble, Vanessa Northington. 1995. *Making a Place for Ourselves: The Black Hospital Movement, 1920- 1945*. New York, NY: Oxford University Press.
- Gamble, Vanessa Northington. 1997. "Black Autonomy versus White Control: Black Hospitals and the Dilemmas of White Philanthropy, 1920-1940." *Minerva* 35 (3): 247–67.
- Germack, Hayley Drew, Ryan Kandrack, and Grant Martsoff. 2019. "When Rural Hospitals Close, the Physician Workforce Goes." *Health Affairs* 38 (12): 2086–94.
- Goldstein, Joshua R., Monica Alexander, Casey Breen, Andrea Miranda González, Felipe Menares, Maria Osborne, Mallika Snyder, and Ugur Yildirim. 2021. "CenSoc Mortality File: Version 2.0."
- Goodman-Bacon, Andrew. 2018. "Public Insurance and Mortality: Evidence from Medicaid Implementation." *Journal of Political Economy* 126 (1): 216–62.
- Goodman-Bacon, Andrew. 2021a. "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics* 225 (2): 254–77.
- Goodman-Bacon, Andrew. 2021b. "The Long-run Effects of Childhood Insurance Coverage: Medicaid Implementation, Adult Health and Labor Market Outcomes." *American Economic Review* 111 (8): 2550–93.
- Haines, Michael. 2008. "Fertility and Mortality in the United States." EH.Net Encyclopedia. <https://eh.net/encyclopedia/fertility-and-mortality-in-the-united-states/> (accessed April 15, 2023).
- Health Insurance Institute. 1977. *Source Book of Health Insurance Data, 1976–1977*. Washington, DC: US Department of Health, Education, and Welfare.
- Hjort, Jonas, Mikkel Sølvsten, and Miriam Wüst. 2017. "Universal Investment in Infants and Long-run Health: Evidence from Denmark's 1937 Home Visiting Program." *American Economic Journal: Applied Economics* 9 (4): 78–104.
- Hoehn-Velasco, Lauren. 2018. "Explaining Declines in US Rural Mortality, 1910-1933: The Role of County Health Departments." *Explorations in Economic History* 70: 42–72.
- Hoehn-Velasco, Lauren. 2021. "The Long-term Impact of Preventative Public Health Programs." *Economic Journal* 131 (634): 797–826.
- Hollingsworth, Alex, Krzysztof Karbownik, Melissa A. Thomasson, and Anthony Wray. 2024. "Replication data for: The Gift of a Lifetime: The Hospital, Modern Medicine, and Mortality." American Economic Association [publisher], Inter-university Consortium for Political and Social Research [distributor]. <https://doi.org/10.3886/E197844V1>.
- IPUMS USA. 2015. "ICPSR County Codes." IPUMS USA. <https://usa.ipums.org/usa/volii/ICPSR.shtml> (accessed August 28, 2023).
- IPUMS USA. 2021. "PLACENHG Code List." IPUMS USA. https://usa.ipums.org/usa/resources/volii/ipums_placenhg_code_list.xlsx (accessed September 5, 2023).

- Jayachandran, Seema, Adriana Lleras-Muney, and Kimberly Smith. 2010. "Modern Medicine and the Twentieth Century Decline in Mortality: Evidence on the Impact of Sulfa Drugs." *American Economic Journal: Applied Economics* 2 (2): 118–46.
- Kozhimannil, Katy, Peiyin Hung, Carrie Henning-Smith, Michelle Casey, and Shailendra Prasad. 2018. "Association Between Loss of Hospital-Based Obstetric Services and Birth Outcomes in Rural Counties in the United States." *JAMA* 319 (12): 1239–47.
- Lazuka, Volha. 2020. "Infant Health and Later-life Labor Market Outcomes: Evidence from the Introduction of Sulfa Antibiotics in Sweden." *Journal of Human Resources* 55 (2): 660–98.
- Lechner, Michael. 2011. "The Estimation of Causal Effects by Difference-in-Difference Methods." *Foundations and Trends in Econometrics* 4 (3): 165–224.
- Lee, David S., Justin McCrary, Marcelo J. Moreira, and Jack Porter. 2022. "Valid t-Ratio Inference for IV." *American Economic Review* 112 (10): 3260–90.
- Manson, Steven, Jonathan Schroeder, David Van Riper, Tracy Kugler, and Steven Ruggles. 2022. "IPUMS National Historical Geographic Information System: Version 17.0 [Dataset]." IPUMS.
- Moehling, Carolyn M., Gregory T. Niemesh, Melissa A. Thomasson, and Jaret Treber. 2020. "Medical Education Reforms and the Origins of the Rural Physician Shortage." *Cliometrica* 14 (2): 181–225.
- Moehling, Carolyn, and Melissa Thomasson. 2014. "Saving Babies: The Impact of Public Education Programs on Infant Mortality." *Demography* 51 (2): 367–86.
- Mullahy, John, and Edward C. Norton. 2022. "Why Transform Y? A Critical Assessment of Dependent-Variable Transformations in Regression Models for Skewed and Sometimes-Zero Outcomes." NBER Working Paper 30735.
- National Center for Health Statistics. 2017. "Health, United States, 2017: Table 015." National Center for Health Statistics.
- North Carolina Bureau of Vital Statistics. 1922. *Annual report of the Bureau of Vital Statistics of the North Carolina State Board of Health, 1922-1948*. Raleigh, NC: Bureau of Vital Statistics of the North Carolina State Board of Health.
- North Carolina Bureau of Vital Statistics. 1933. *Annual report of the Bureau of Vital Statistics of the North Carolina State Board of Health, 1933*. Raleigh, NC: Bureau of Vital Statistics of the North Carolina State Board of Health.
- Pei, Zhuan, Jörn-Steffen Pischke, and Hannes Schwandt. 2019. "Poorly Measured Confounders Are More Useful on the Left than on the Right." *Journal of Business and Economic Statistics* 37 (2): 205–16.
- Pollitt, Phoebe Ann. 2017. *African American Hospitals in North Carolina: 39 Institutional Histories, 1880-1967*. Jefferson, NC: McFarland.
- Roth, Jonathan, Pedro Sant'Anna, Alyssa Bilinski, and John Poe. 2022. "What's Trending in Difference-in-Differences? A Synthesis of the Recent Econometrics Literature." Unpublished.
- Santos Silva, J. M. C., and Silvana Tenreyro. 2011. "Further Simulation Evidence on the Performance of the Poisson Pseudo-maximum Likelihood Estimator." *Economics Letters* 112 (2): 220–22.
- Soares, Rodrigo R. 2007. "On the Determinants of Mortality Reductions in the Developing World." *Population and Development Review* 33 (2): 247–87.
- Social Security Administration. 2007. "Numerical Identification Files (NUMIDENT), created, 1936-2007, documenting the period 1936-2007 [dataset]." Social Security Administration.
- Starr, Paul. 2017. *The Social Transformation of American Medicine*, New York, NY: Basic Books, (Orig. pub. 1982).
- Surveillance, Epidemiology, and End Results Program. 2022. "County-Level Population Files - Single-year Age Groups." <https://seer.cancer.gov/popdata/download.html> (accessed June 29, 2022).
- Thomas, Karen Kruse. 2011. *Deluxe Jim Crow: Civil Rights and American Health Policy, 1935-1954*. Athens, GA: University of Georgia Press.
- Thomasson, Melissa A. 2002. "From sickness to Health: The Twentieth-Century Development of US Health Insurance." *Explorations in Economic History* 39 (3): 233–53.
- Thomasson, Melissa, and Jaret Treber. 2008. "From Home to Hospital: The Evolution of Childbirth in the United States, 1928-1940." *Explorations in Economic History* 45 (1): 76–99.
- US Geological Survey. 2019. *Geographic Names Information System (GNIS)*. Reston, VA: US Geological Survey.
- Viscusi, Kip. 2018. "Best Estimate Selection Bias in the Value of a Statistical Life." *Journal of Benefit-Cost Analysis* 9 (2): 205–46.
- Wooldridge, Jeffrey M. 2021. "Two-Way Fixed Effects, the Two-Way Mundlak Regression, and Difference-in-Differences Estimators." Unpublished.
- Yamey, Gavin, Dean Jamison, Odd Hanssen, and Agnes Soucat. 2019. "Financing Global Common Good for Health: When the World is a Country." *Health Systems and Reform* 5 (3): 334–49.